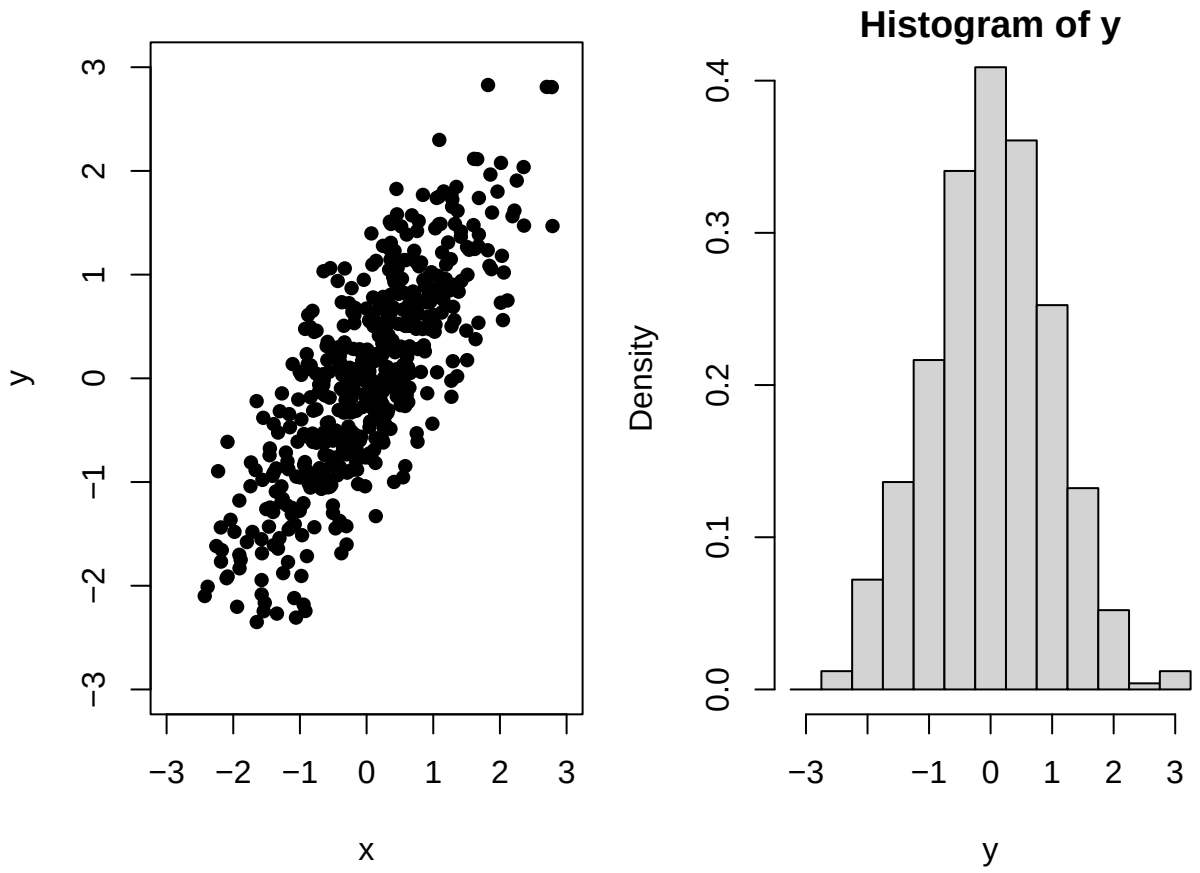
# Solutions to Review Questions for Lecture 8

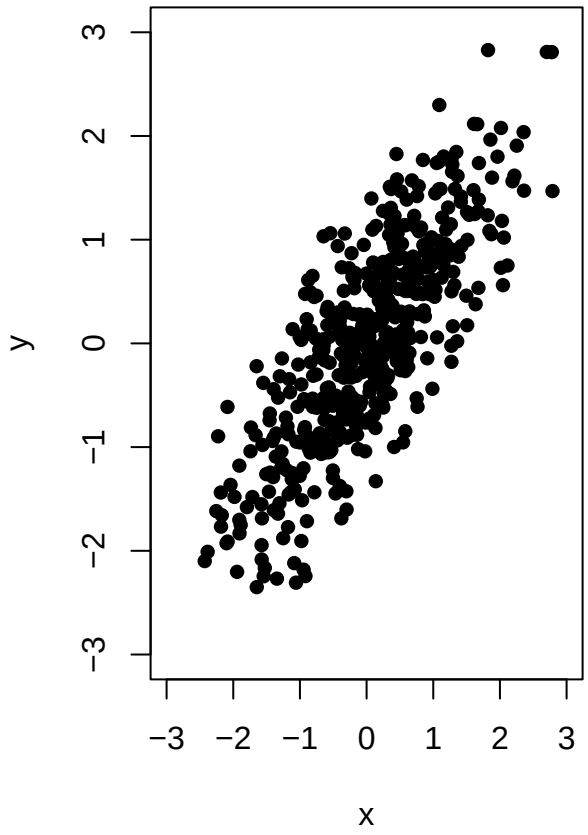
2025-02-13

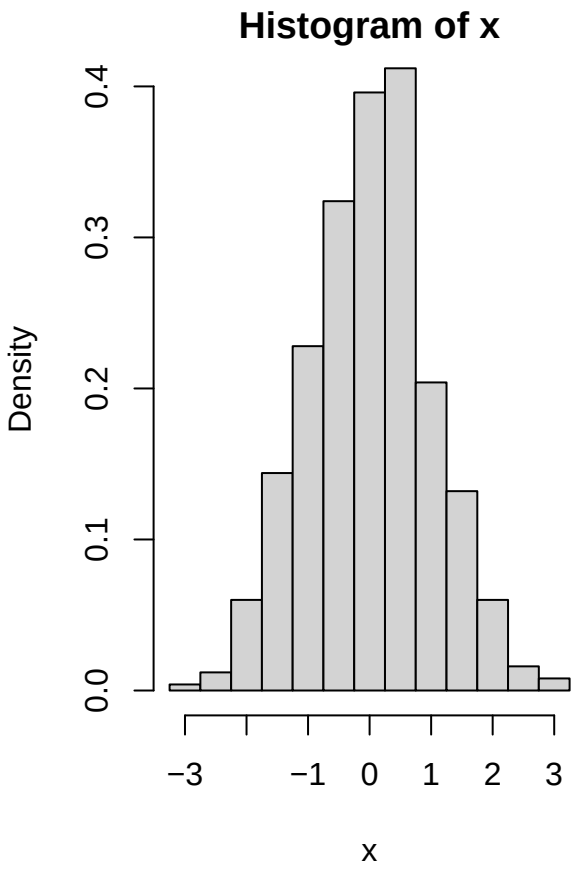
1. The graphs below give the marginal and conditional distributions for plot A

```
marginaly(fb0.8[,1],fb0.8[,2])
```

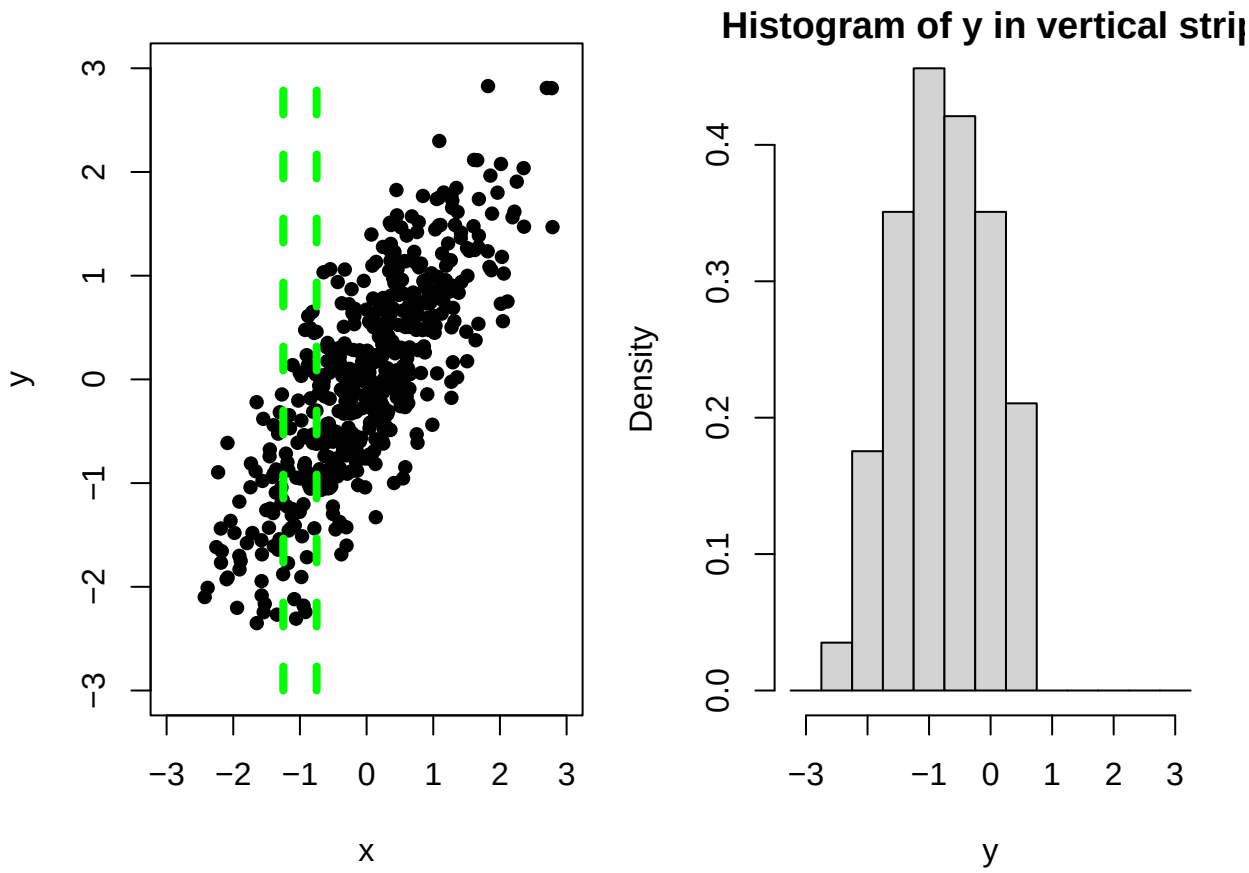



```
marginalx(fb0.8[,1],fb0.8[,2])
```

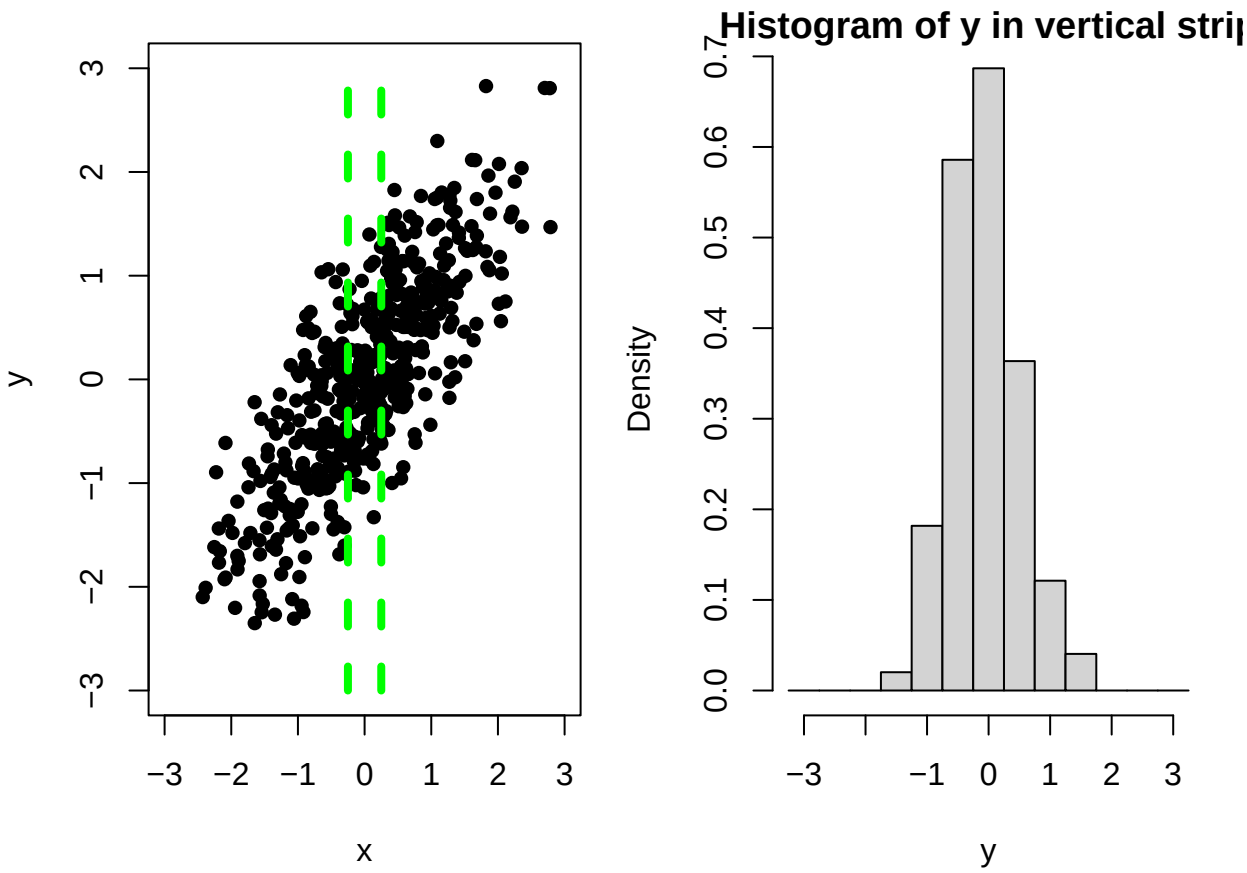



```
conditionaly(fb0.8[,1],fb0.8[,2],-1)
```

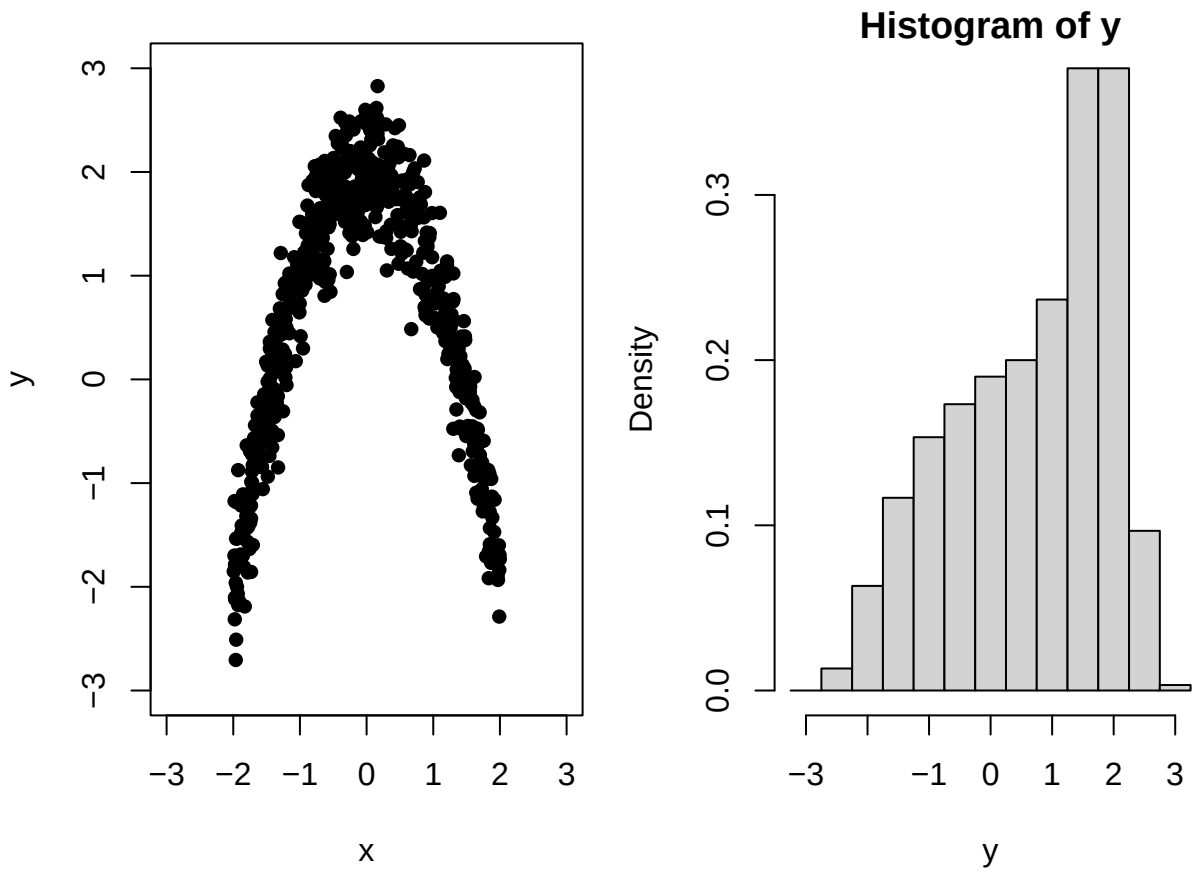



```
conditionaly(fb0.8[,1],fb0.8[,2],0)
```

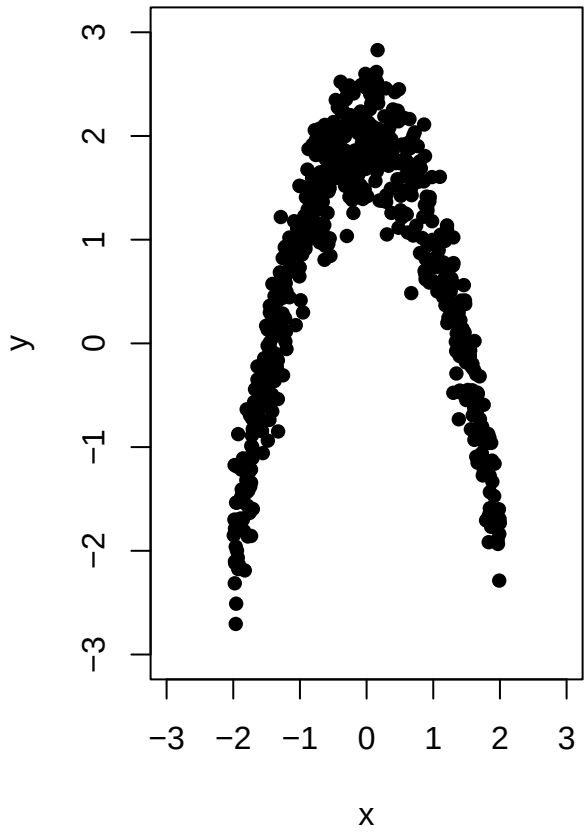


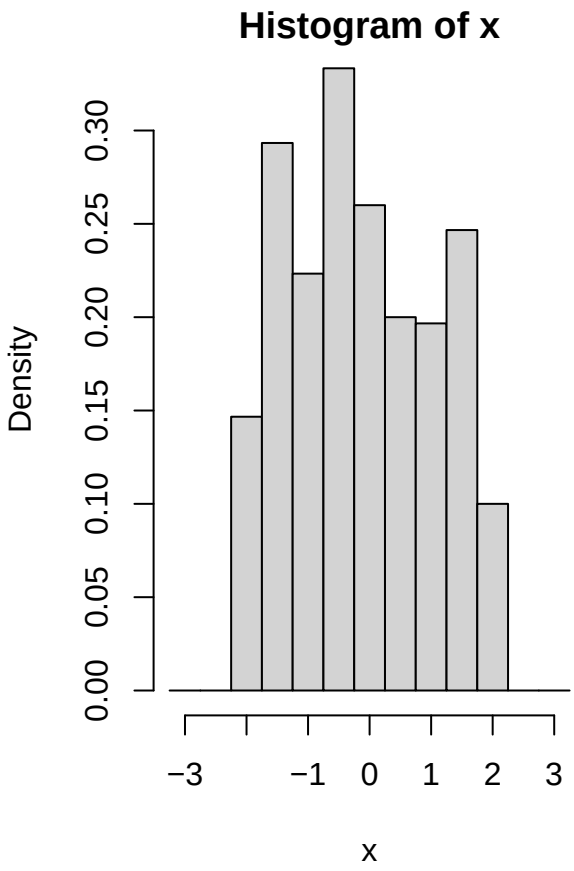

2. The graphs below give the marginal and conditional distributions for plot B

```
marginaly(arc[,1],arc[,2])
```

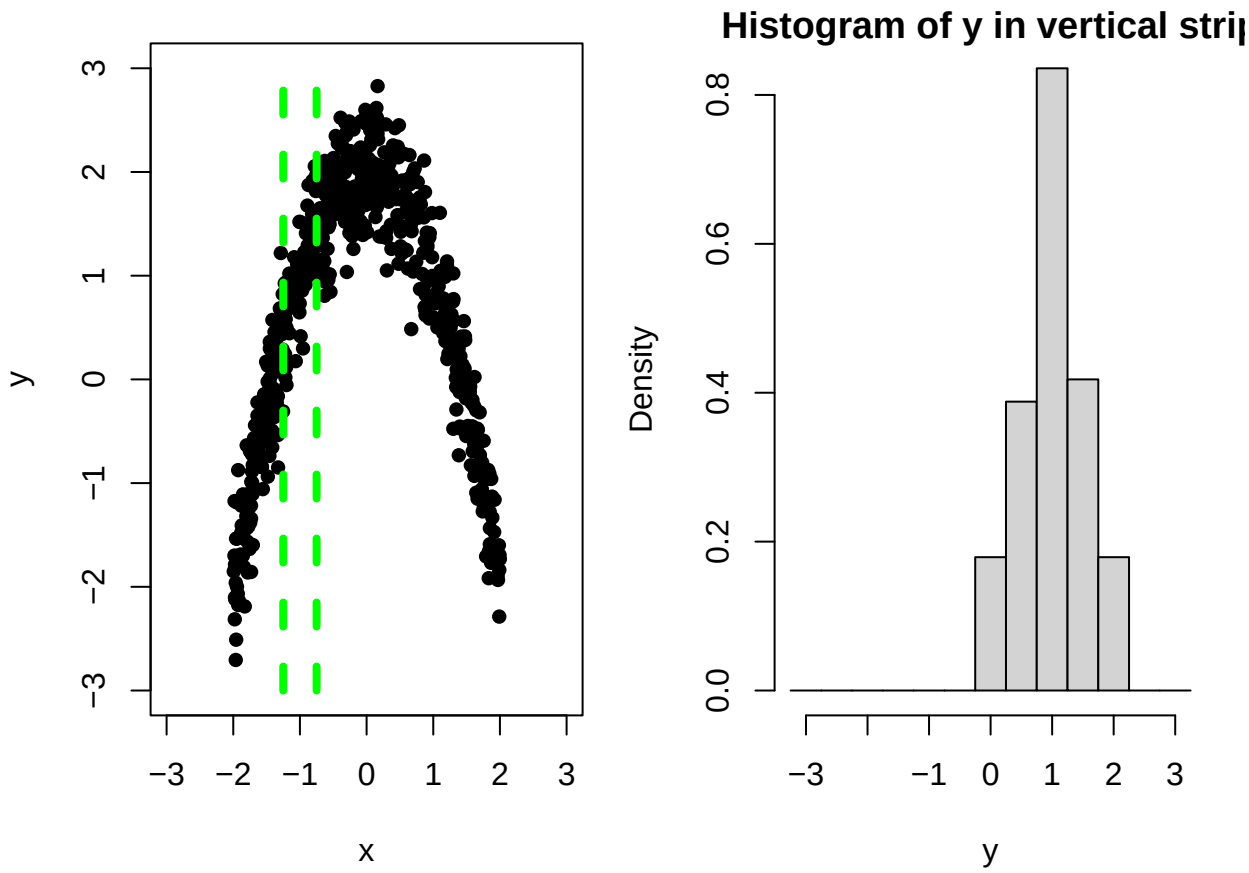



```
marginalx(arc[,1],arc[,2])
```

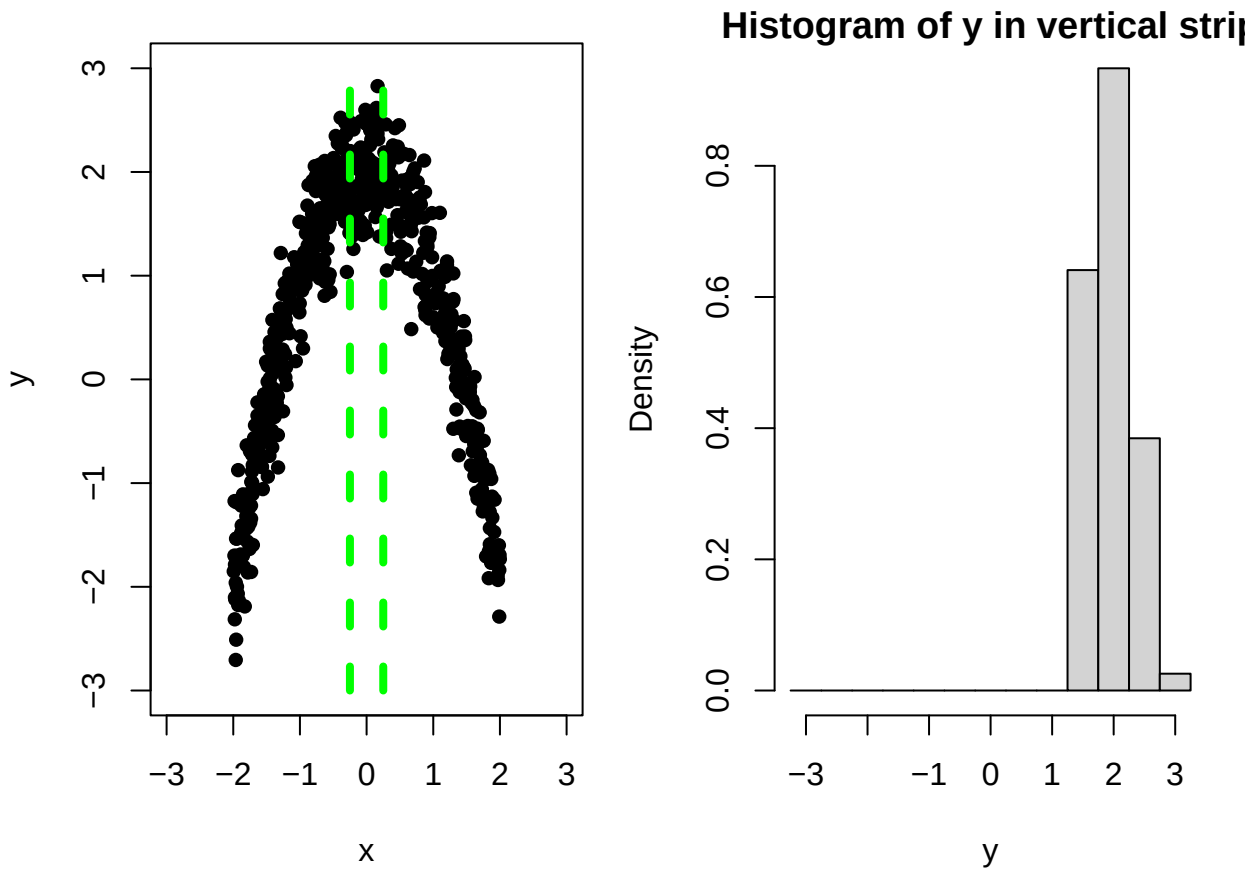


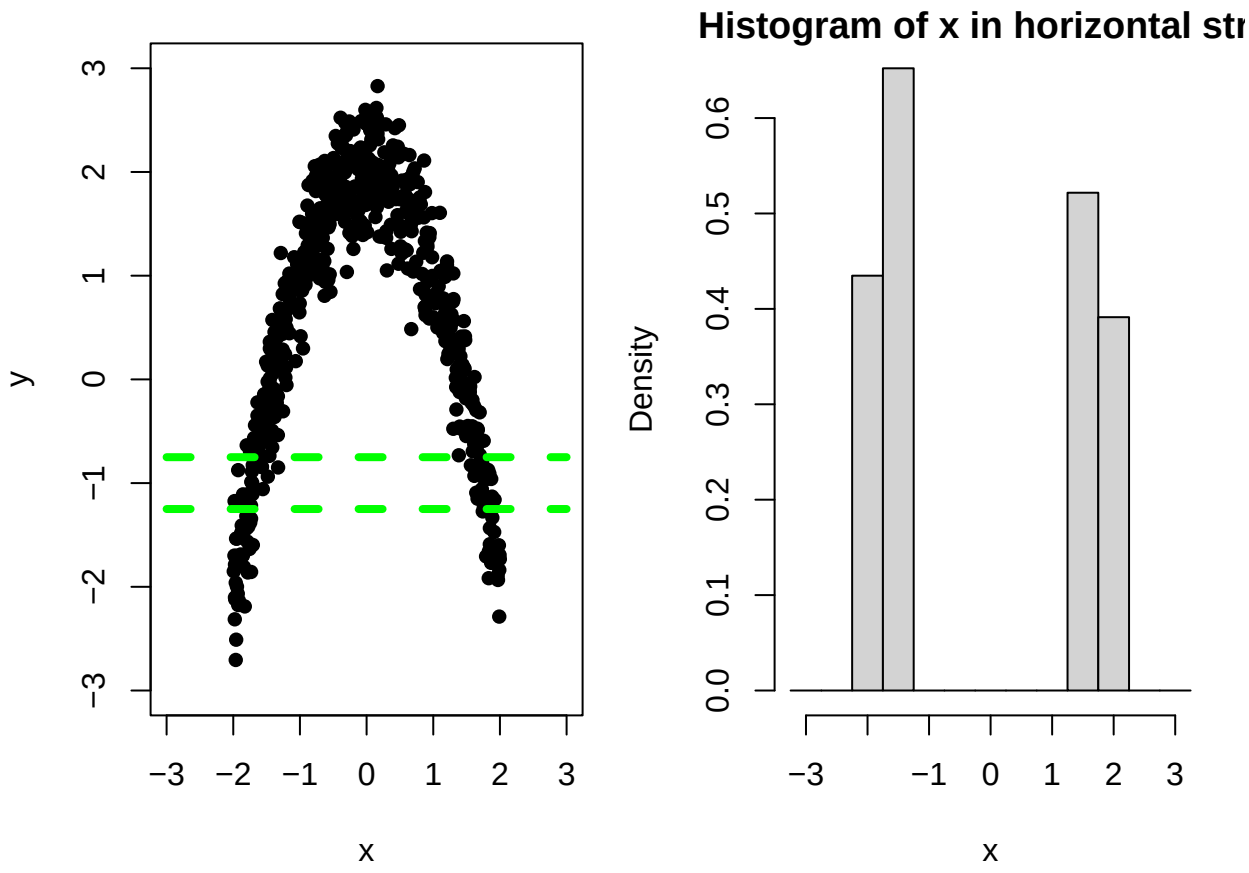

```
conditionaly(arc[,1],arc[,2],-1)
```



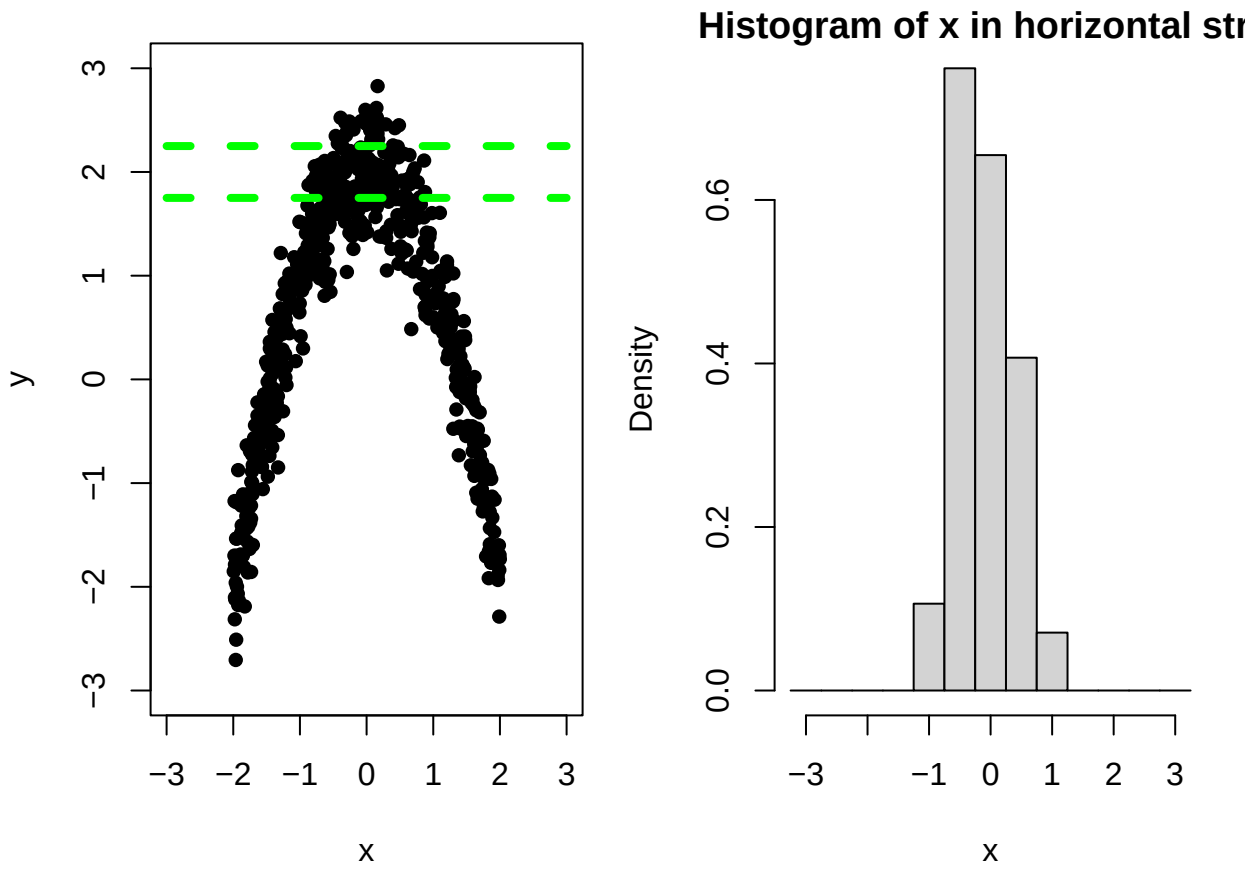

```
conditionaly(arc[,1],arc[,2],0)
```




```
conditionalx(arc[,1],arc[,2],-1)
```

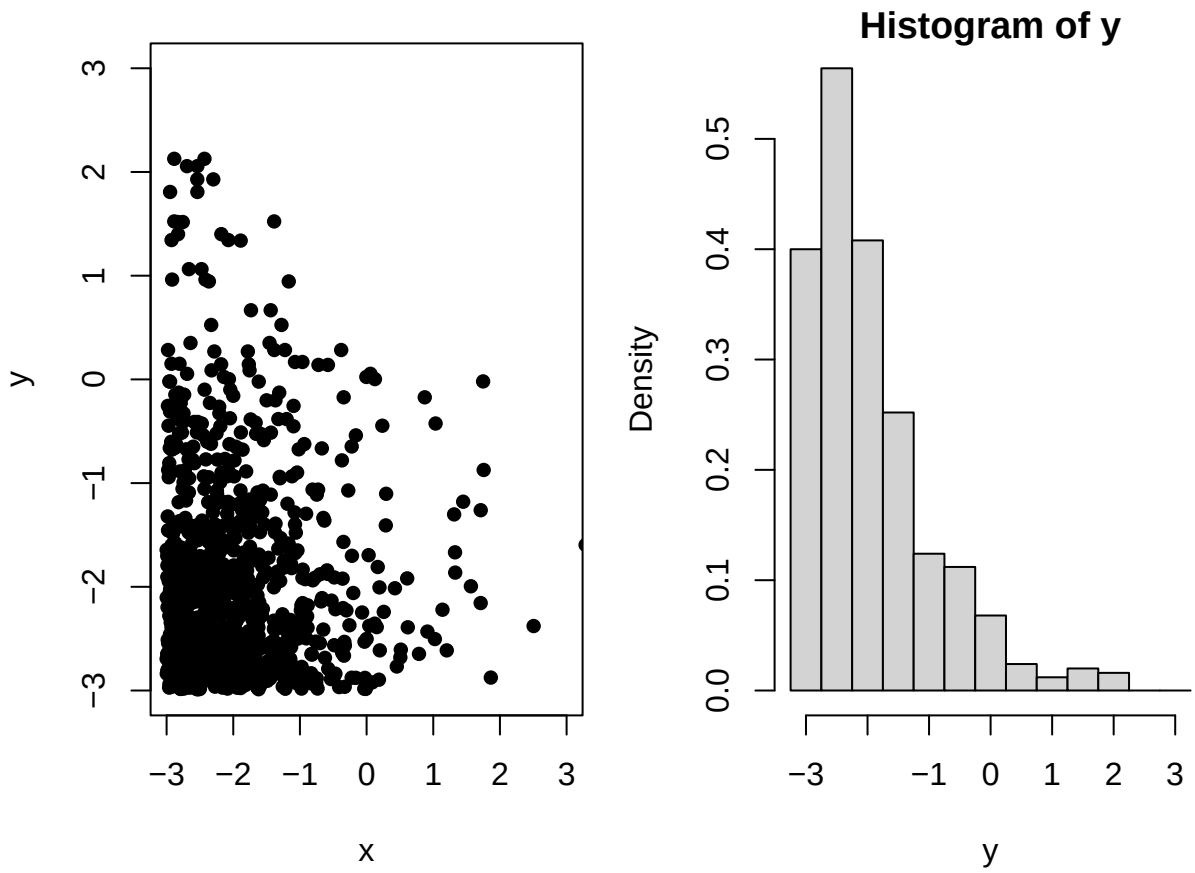



```
conditionalx(arc[,1],arc[,2],2)
```

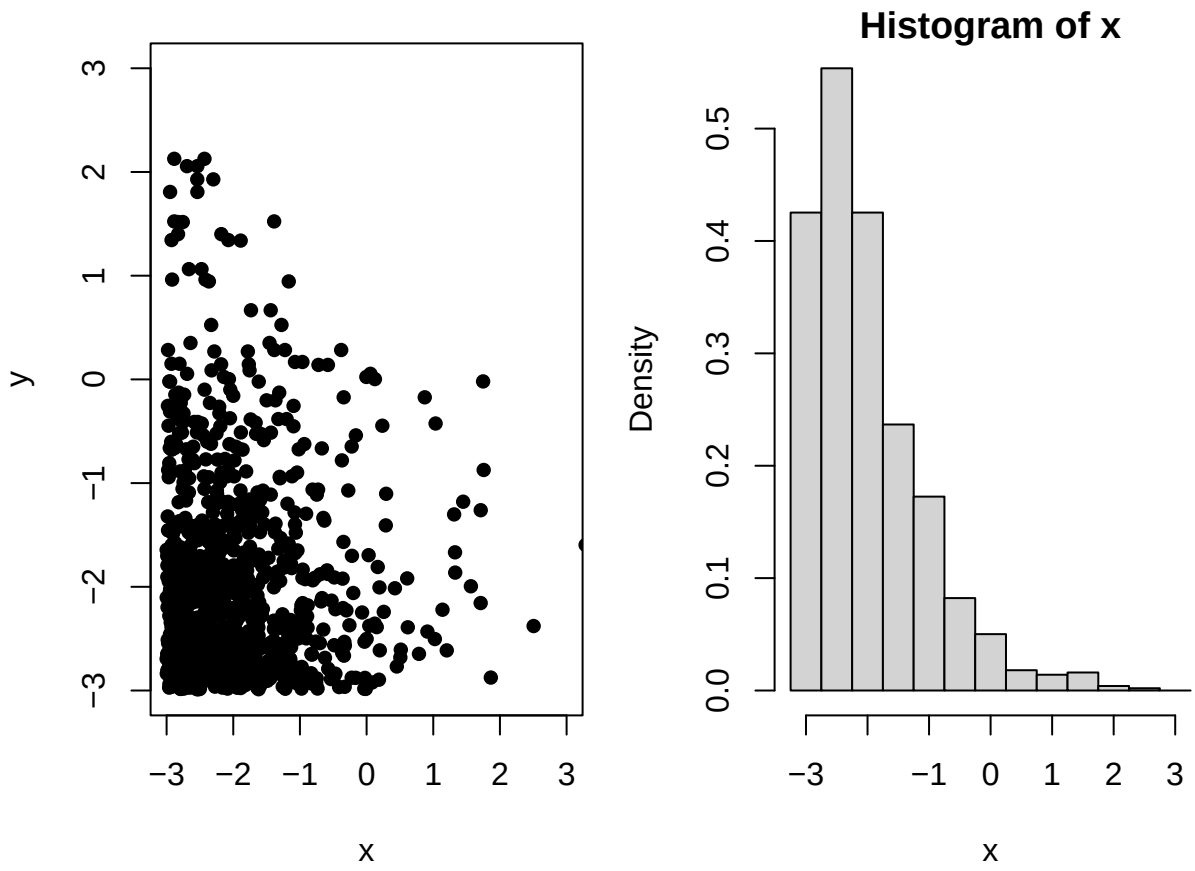



3. The graphs below give the marginal and conditional distributions for plot C

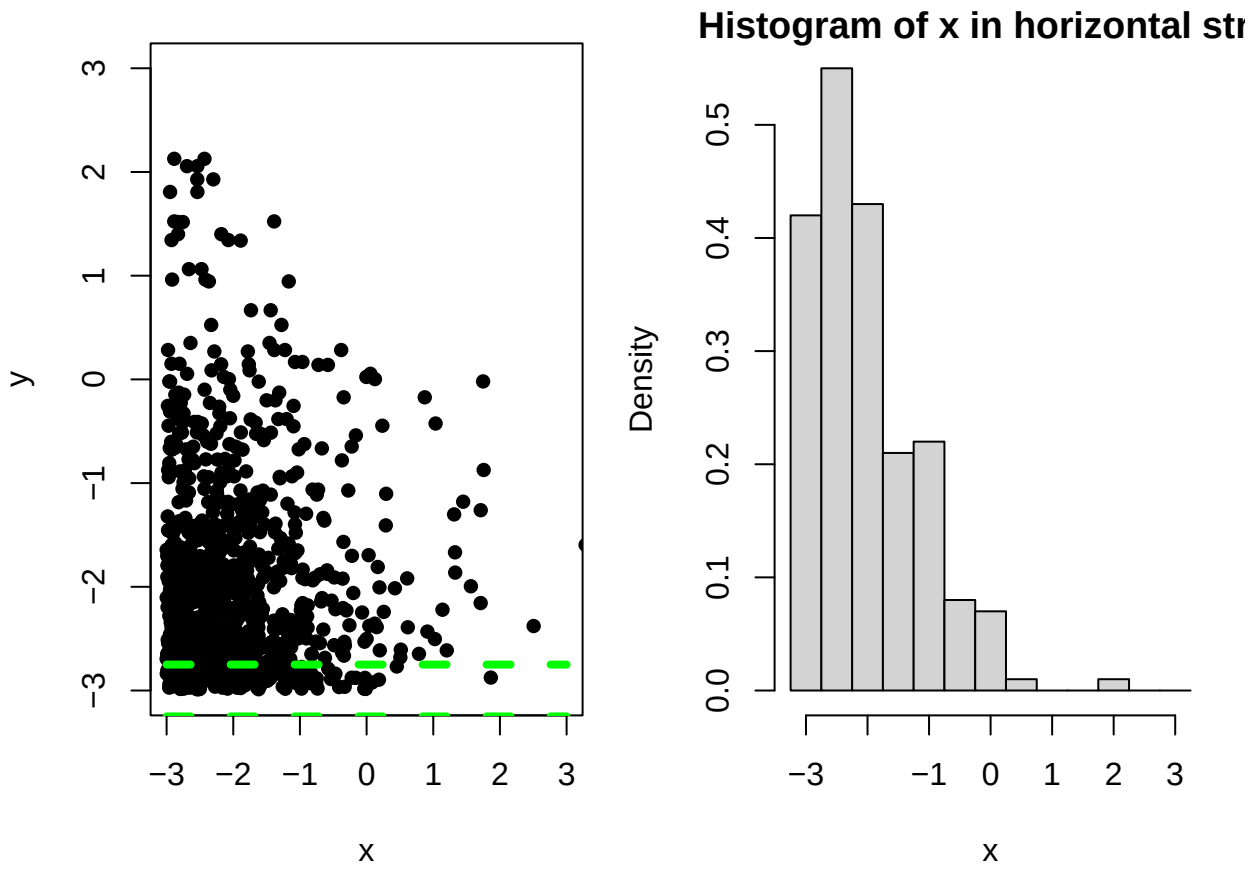
```
marginaly(rtail[,1],rtail[,2])
```



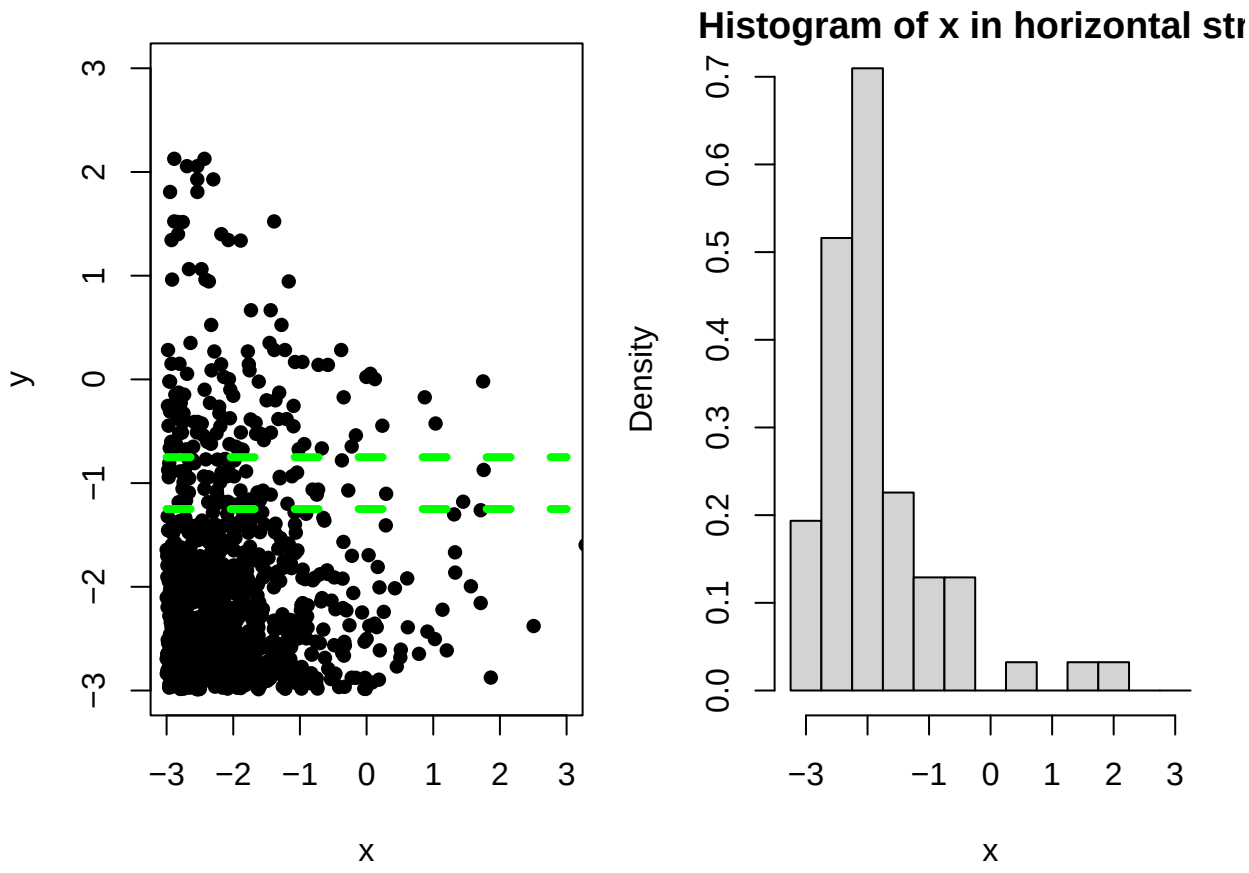

```
marginalx(rtail[,1],rtail[,2])
```




```
conditionalx(rtail[,1],rtail[,2],-3)
```

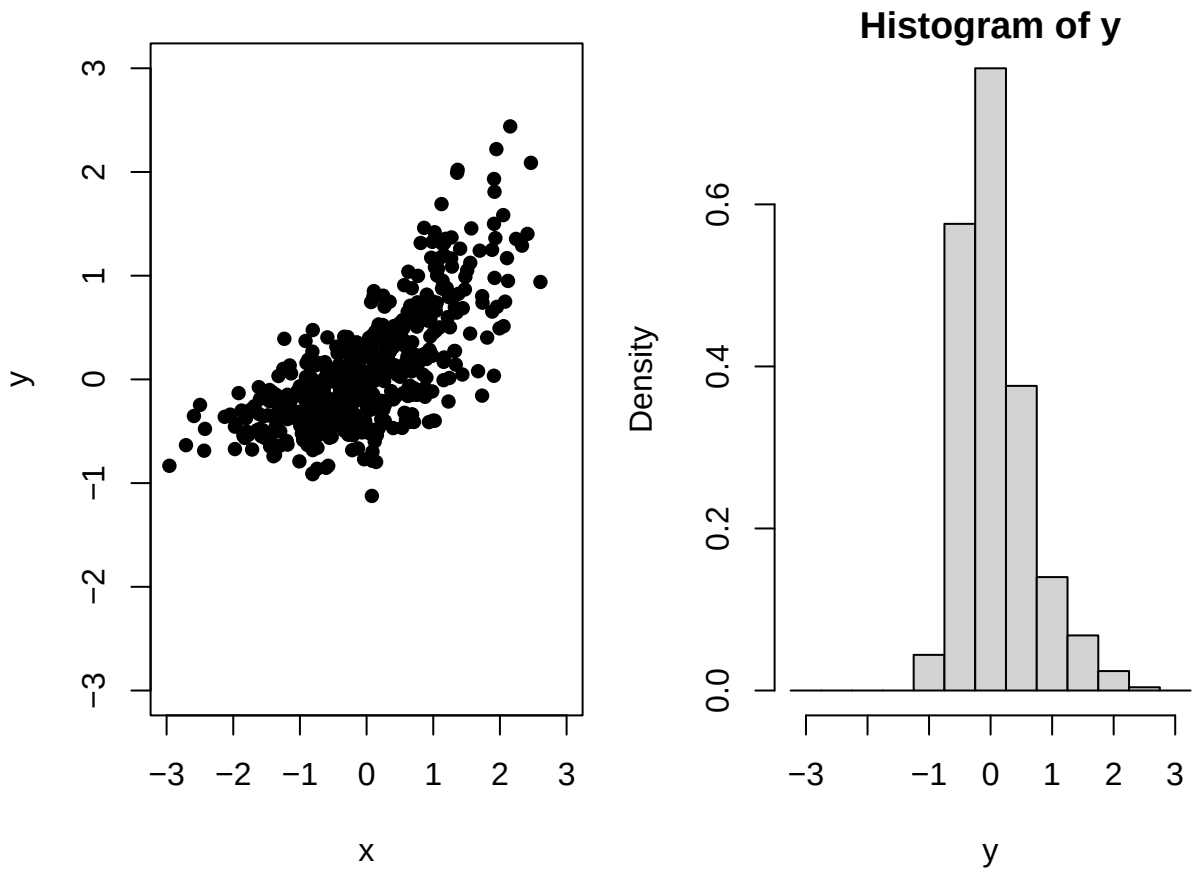



```
conditionalx(rtail[,1],rtail[,2],-1)
```

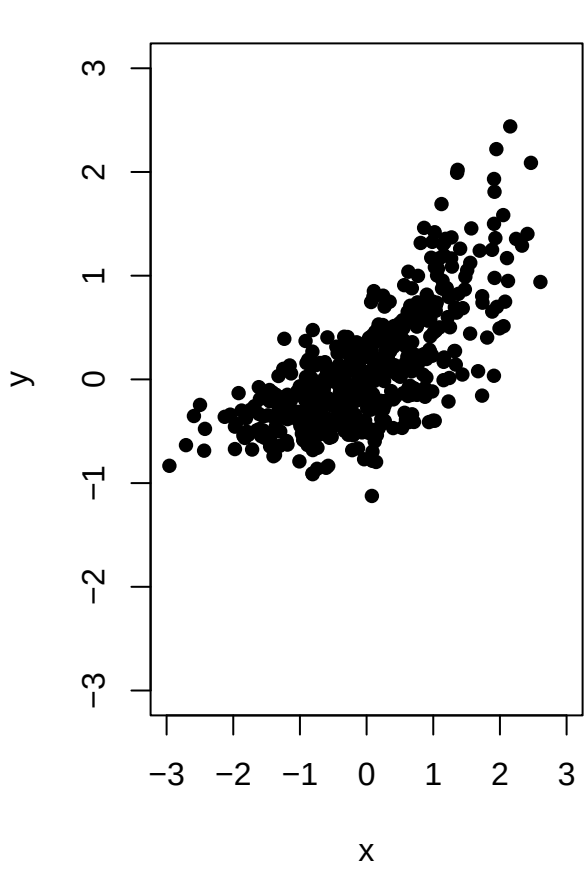


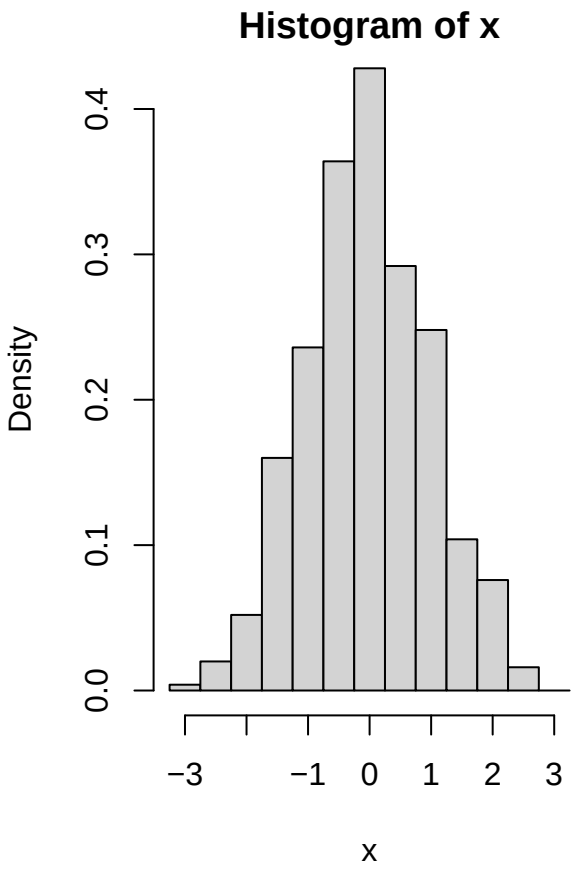

4. The graphs below give the marginal and conditional distributions for plot D

```
marginaly(hetero[,1],hetero[,2])
```

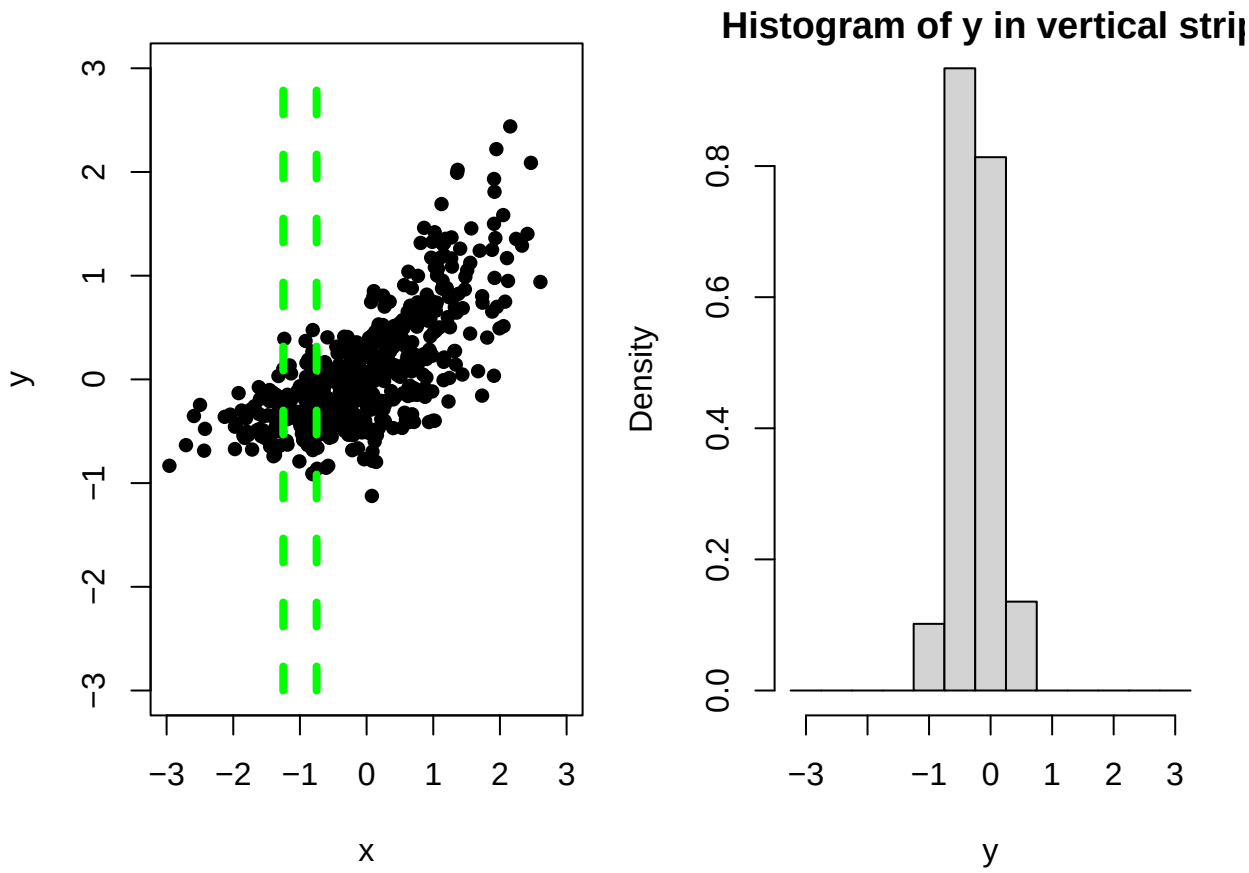



```
marginalx(hetero[,1],hetero[,2])
```

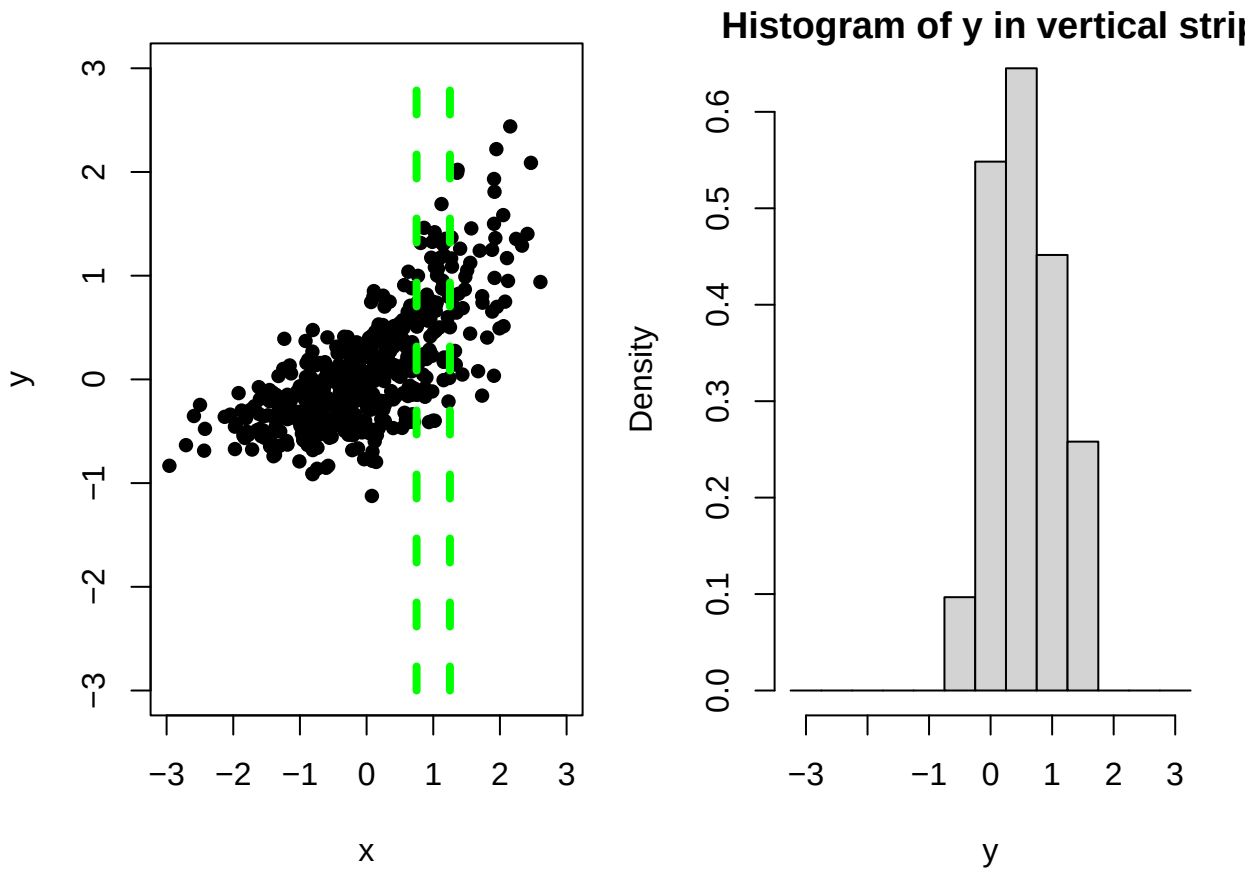



```
conditionaly(hetero[,1],hetero[,2],-1)
```

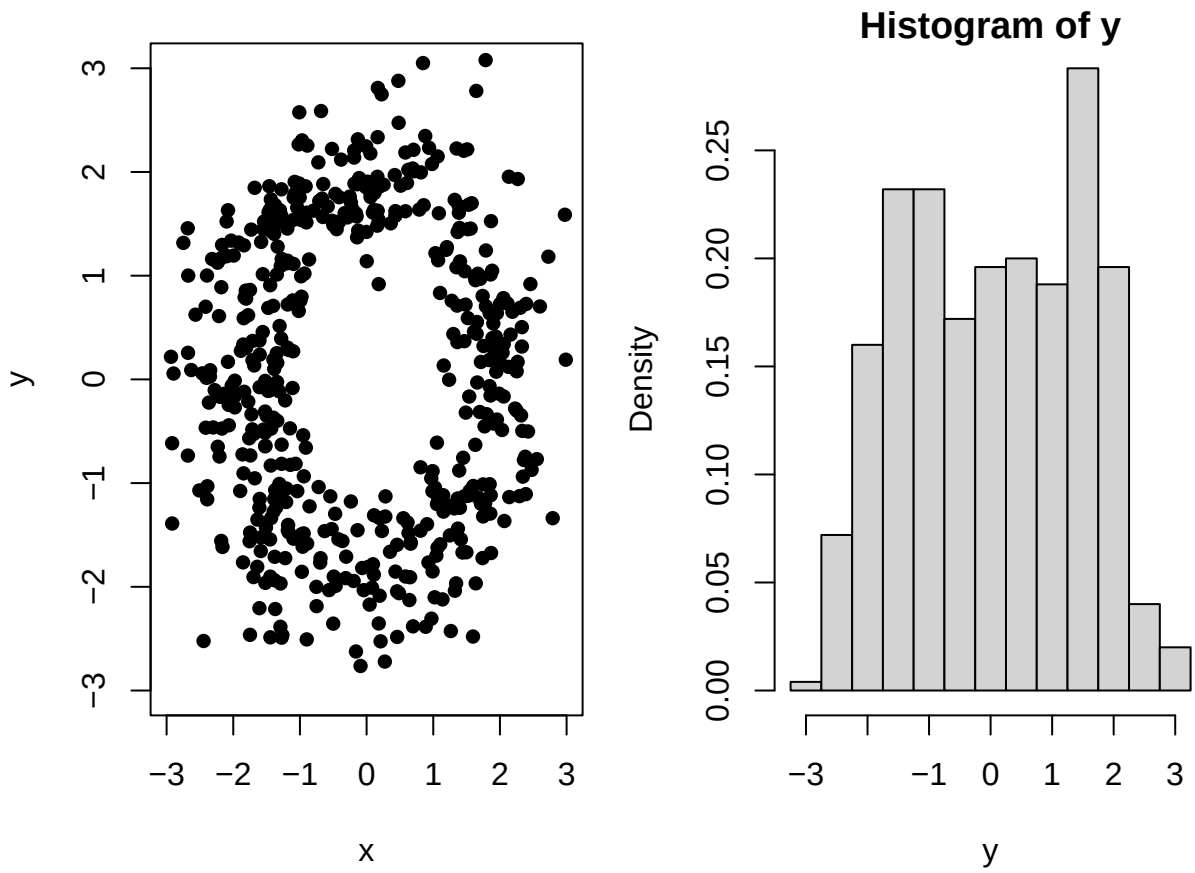



```
conditionaly(hetero[,1],hetero[,2],1)
```

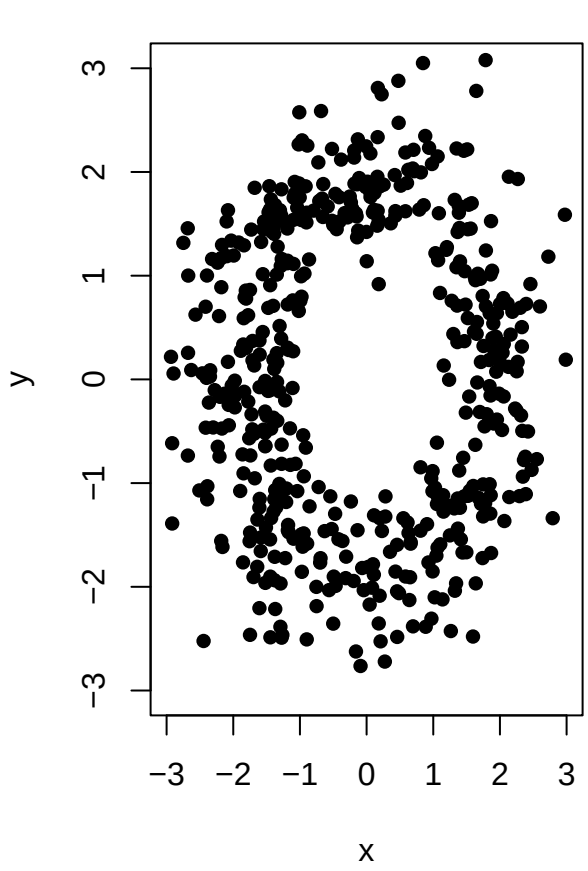


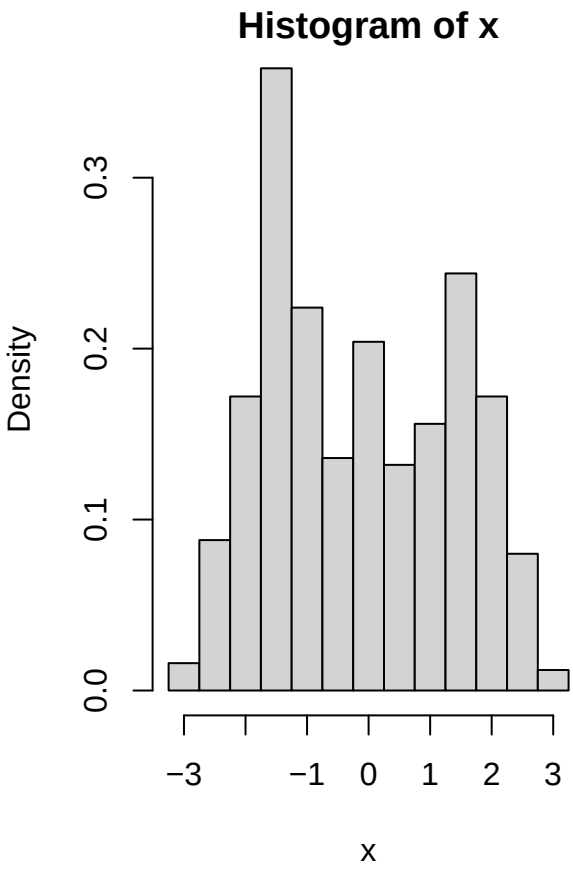

5. The graphs below give the marginal and conditional distributions for plot E

```
marginaly(sinusoid[,1],sinusoid[,2])
```

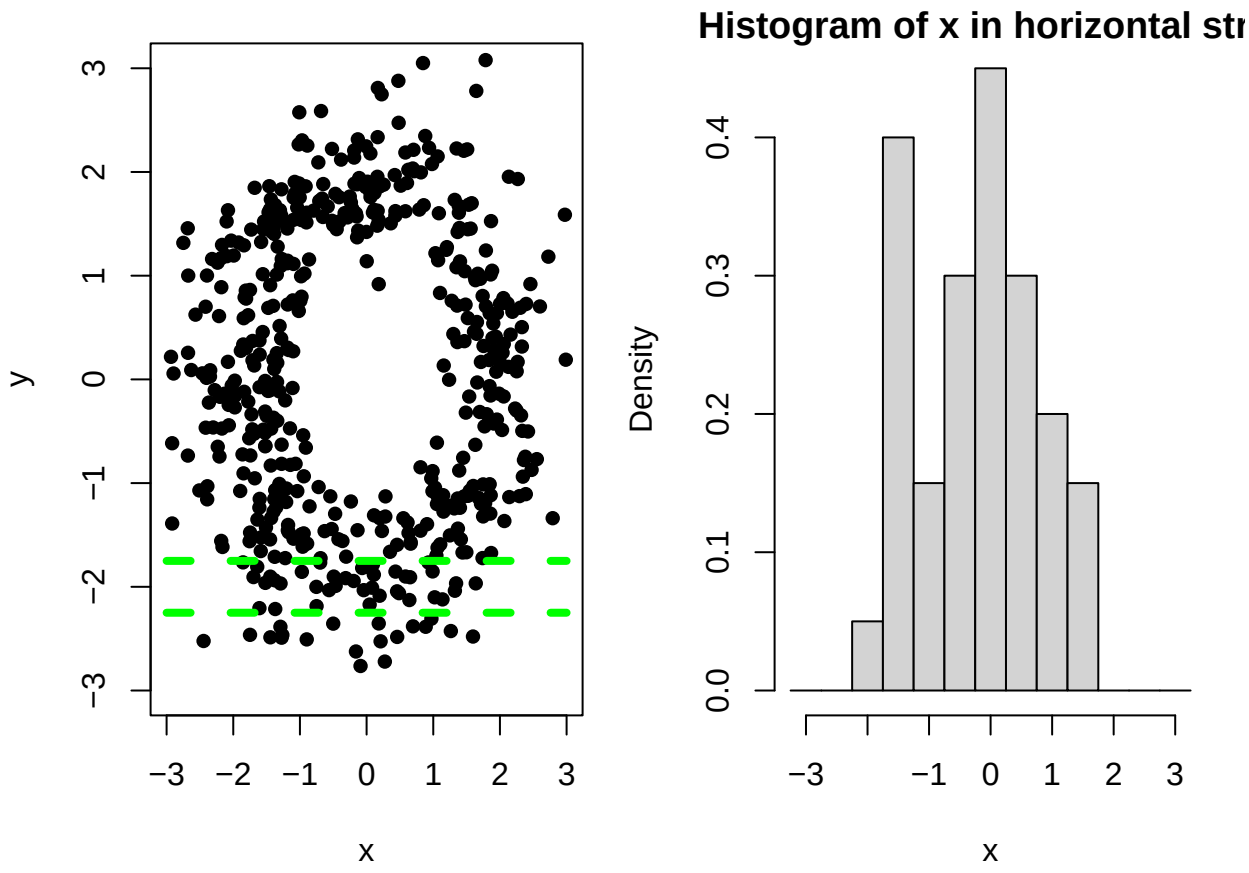



```
marginalx(sinusoid[,1],sinusoid[,2])
```

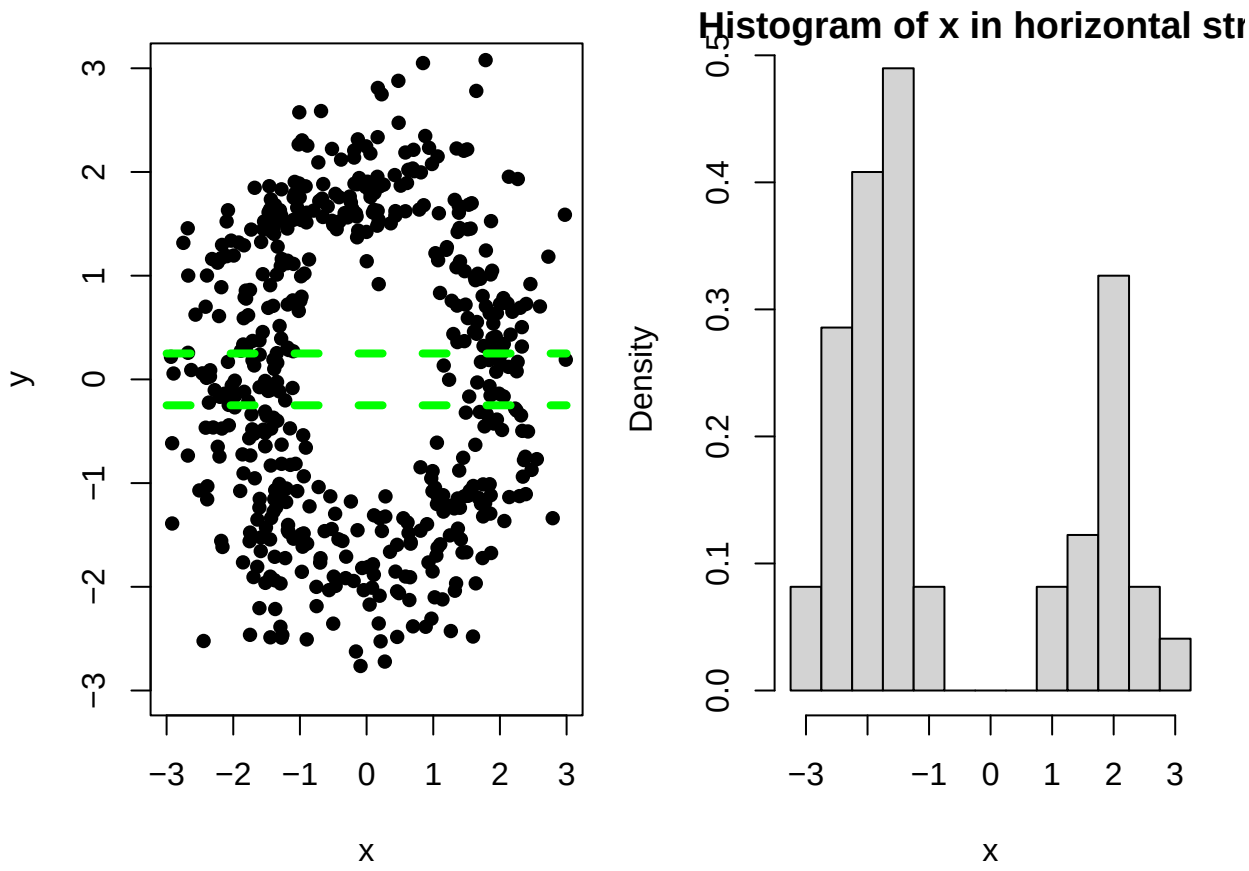



```
conditionalx(sinusoid[,1],sinusoid[,2],-2)
```

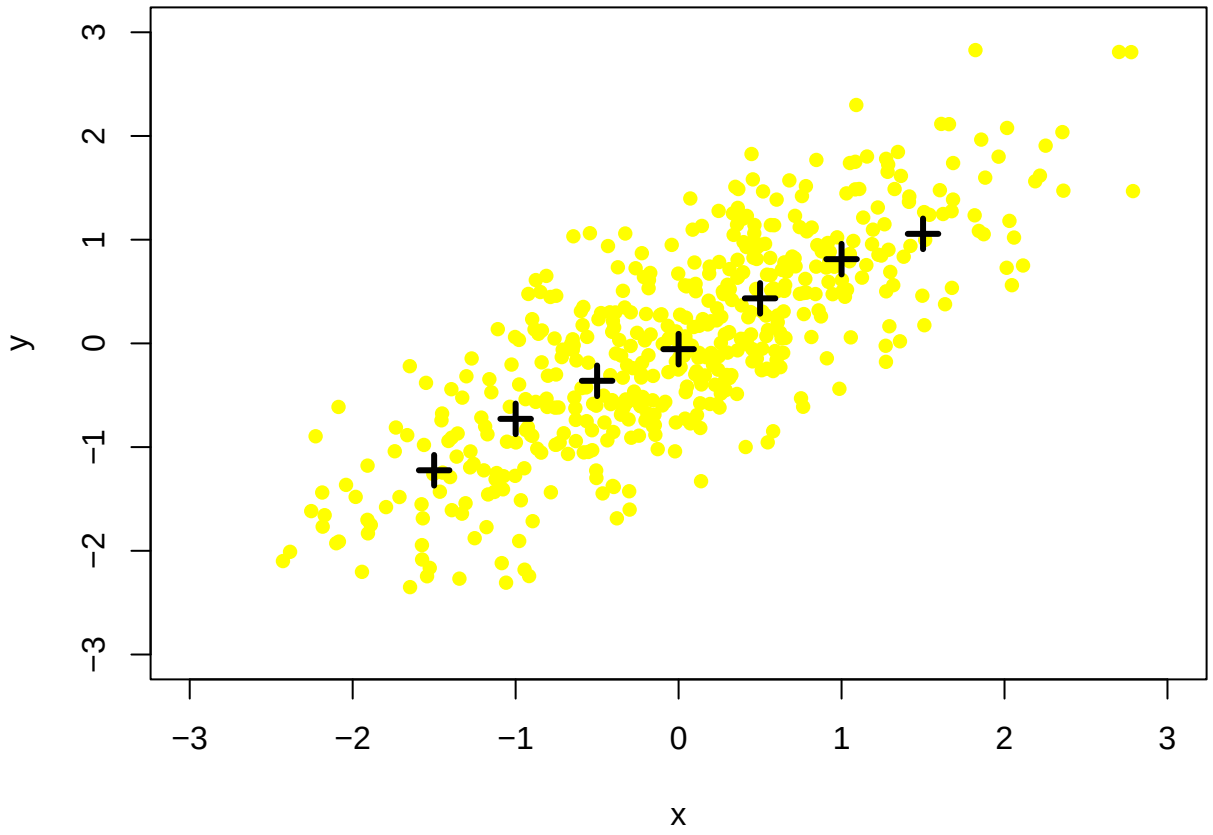



```
conditionalx(sinusoid[,1],sinusoid[,2],0)
```



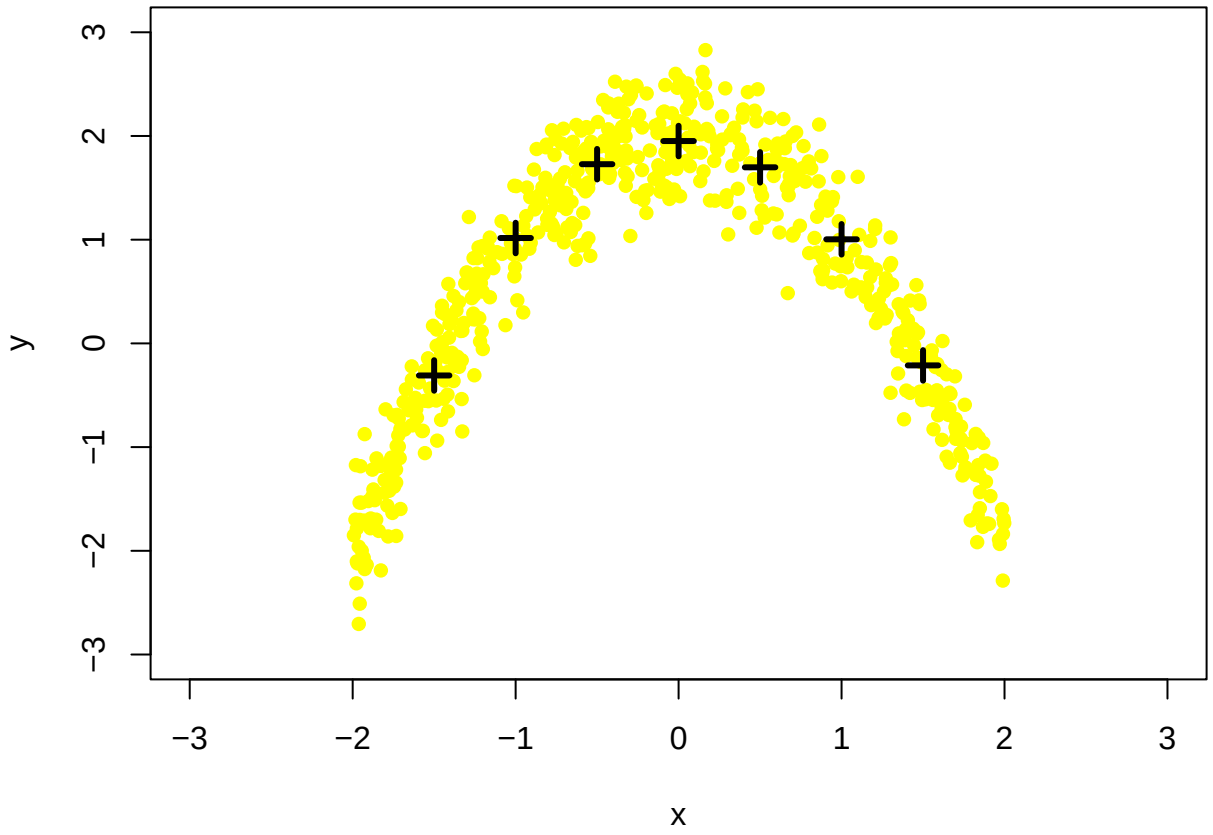

6. $mean(x) \approx 0, mean(y) \approx 0, SD(x) \approx (3 - -3)/4 = 1.5, SD(y) \approx (3 - (-2.5))/4 = 1.4.$
7. $mean(x) \approx 0, mean(y) \approx 0.5, SD(x) \approx (2 - -2)/4 = 1, SD(y) \approx (2.75 - (-2.75))/4 = 1.4.$
8. $mean(x) \approx -1.5, mean(y) \approx -1.5, SD(x) \approx (2.75 - -3)/4 = 1.4, SD(y) \approx (2.25 - (-3))/4 = 1.4.$
9. $mean(x) \approx 0, SD(x) \approx (2.75 - -3)/4 = 1.4, mean(y) \approx 0.5, SD(y) \approx (2 - (-1))/4 = 0.8.$
10. $mean(x) \approx -0.25, SD(x) \approx (3 - -3)/4 = 1.5, mean(y) \approx 0.25, SD(y) \approx (3 - (-3))/4 = 1.5.$
11. The graph below gives the graph of averages for plot A

```
graphavgy(fb0.8[,1],fb0.8[,2])
```



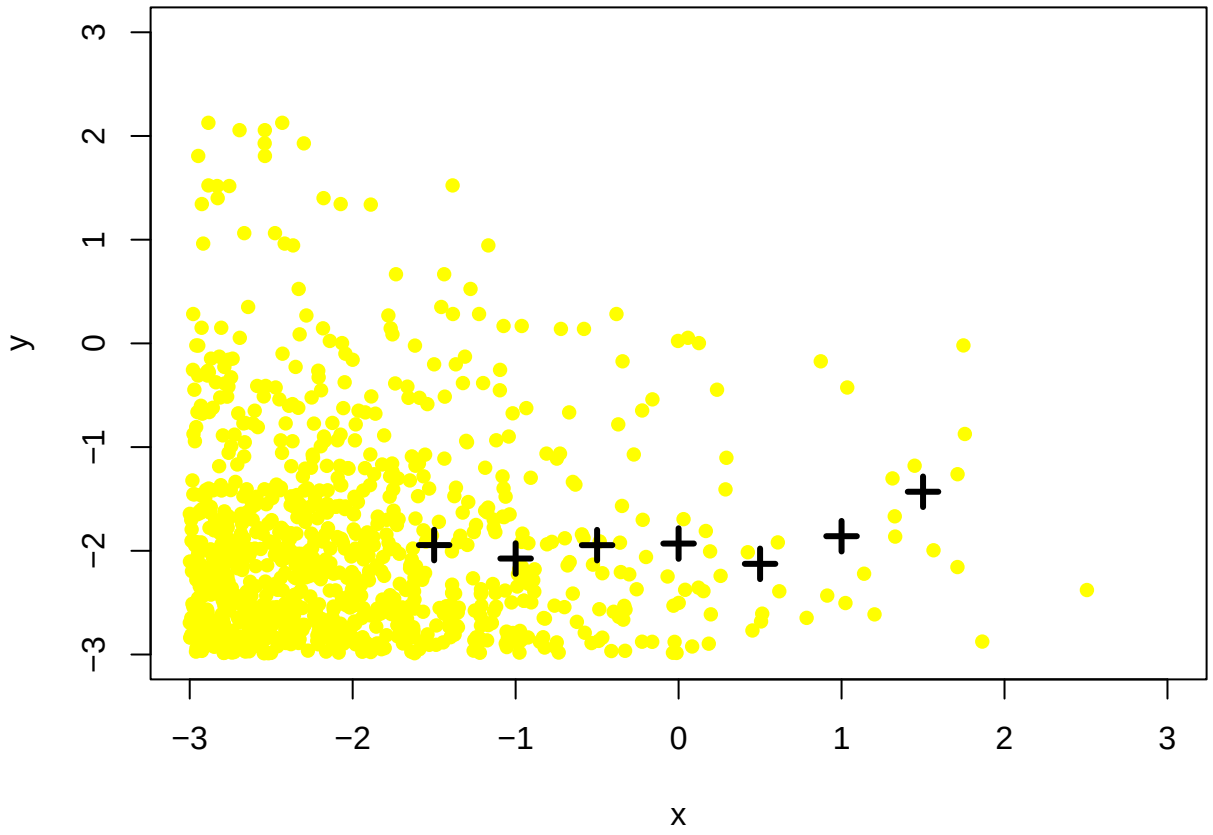

12. The graph below gives the graph of averages for plot B

```
graphavgy(arc[,1],arc[,2])
```



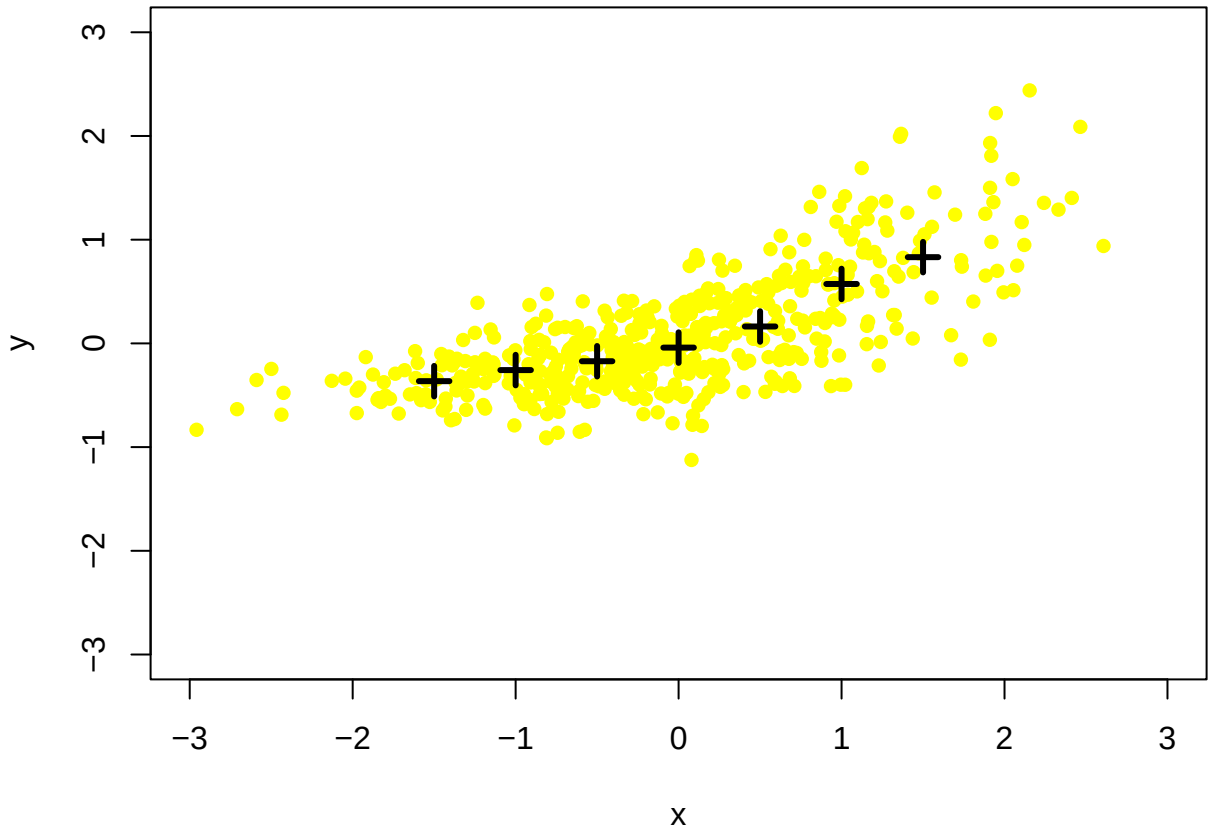

13. The graph below gives the graph of averages for plot C

```
graphavgy(rtail[,1],rtail[,2])
```



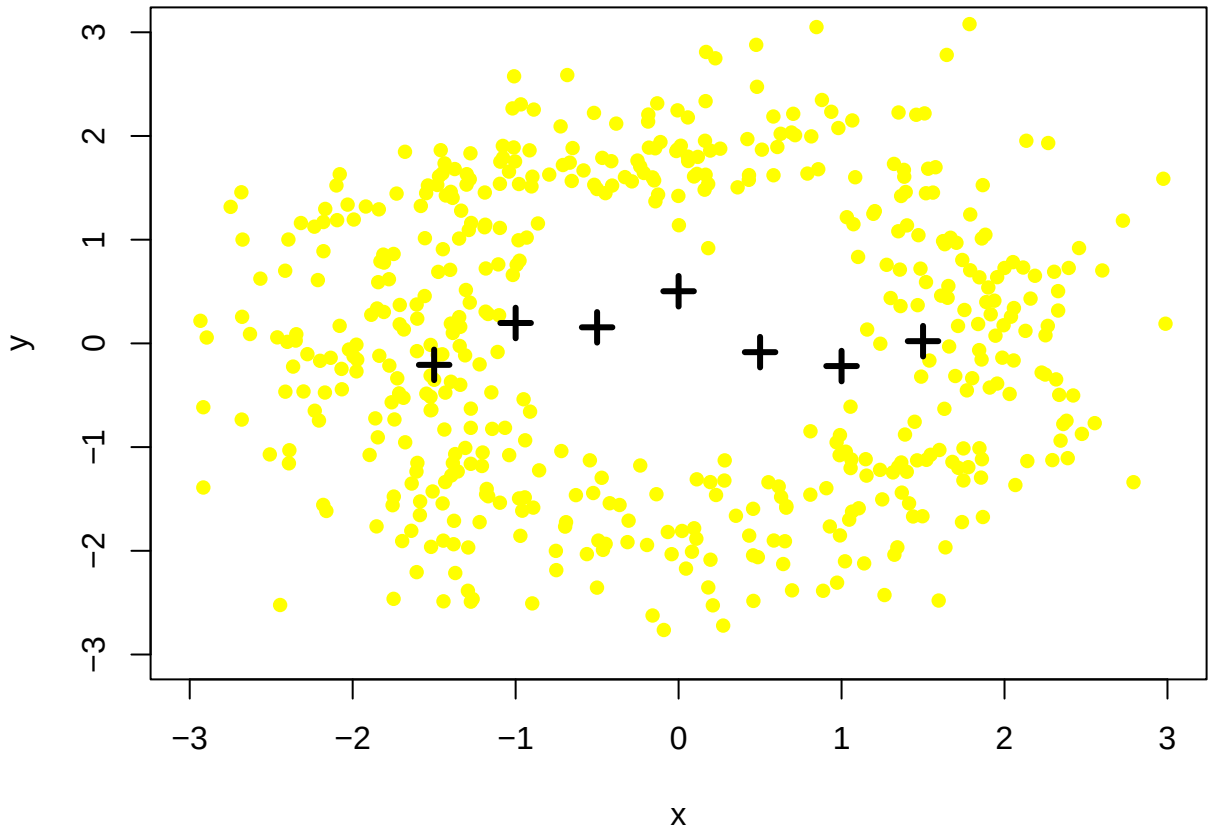

14. The graph below gives the graph of averages for plot D

```
graphavgy(hetero[,1],hetero[,2])
```



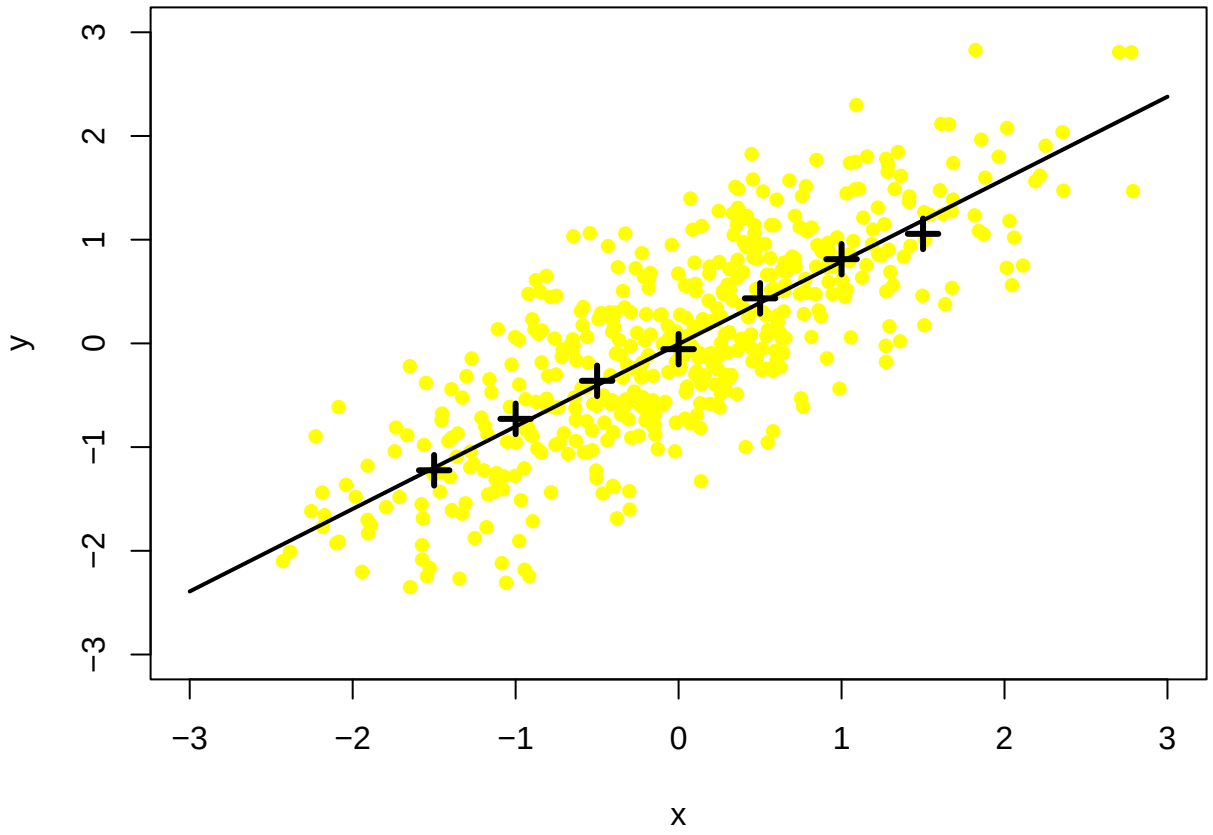

15. The graph below gives the graph of averages for plot E

```
graphavgy(sinusoid[,1],sinusoid[,2])
```



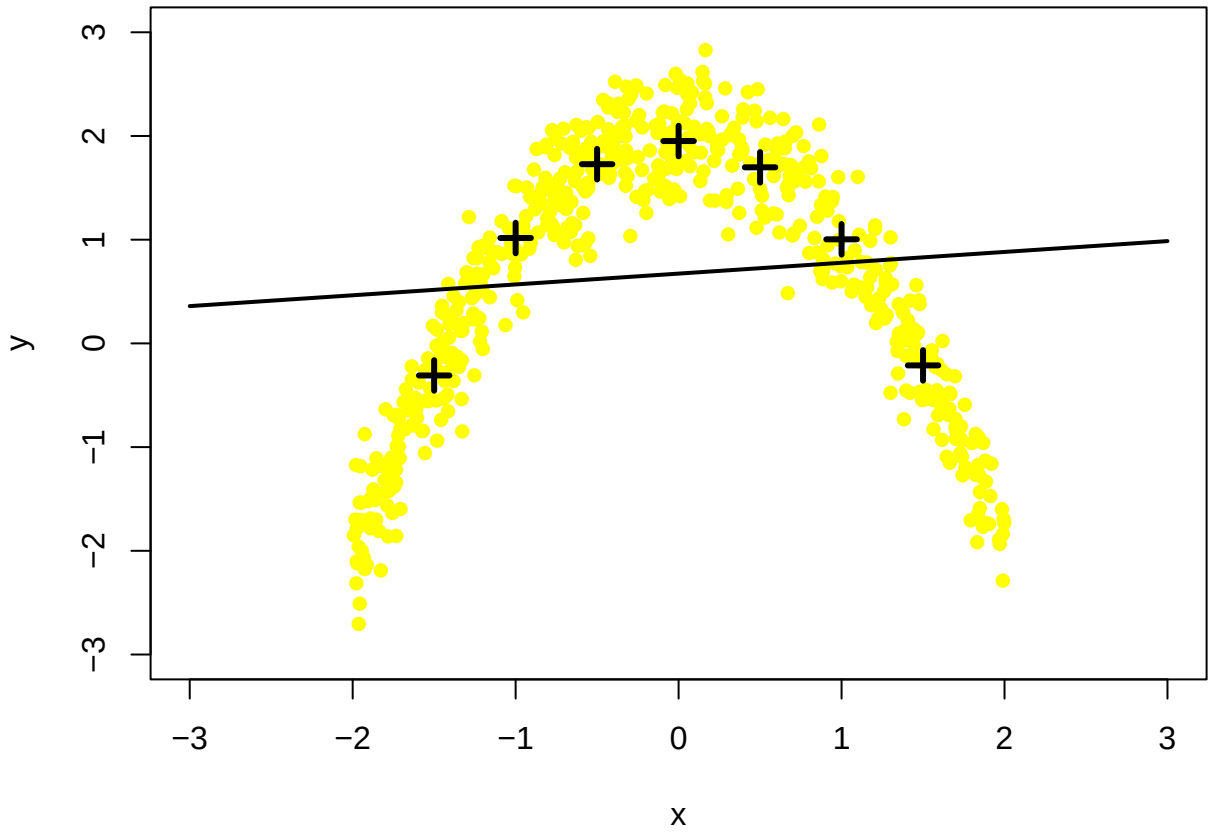

16. The graph below gives the regression line for plot A

```
graphavgregy(fb0.8[,1],fb0.8[,2])
```



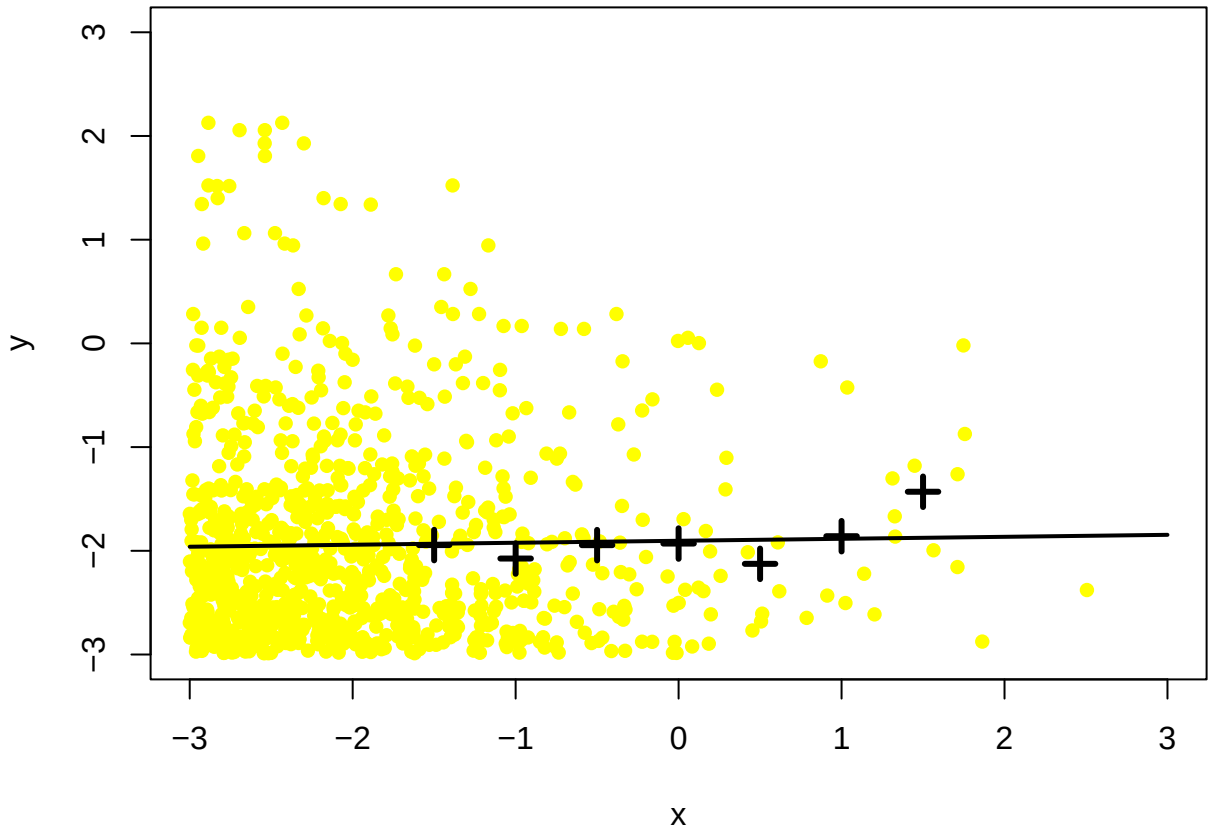

17. The graph below gives the regression line for plot B

```
graphavgregy(arc[,1],arc[,2])
```



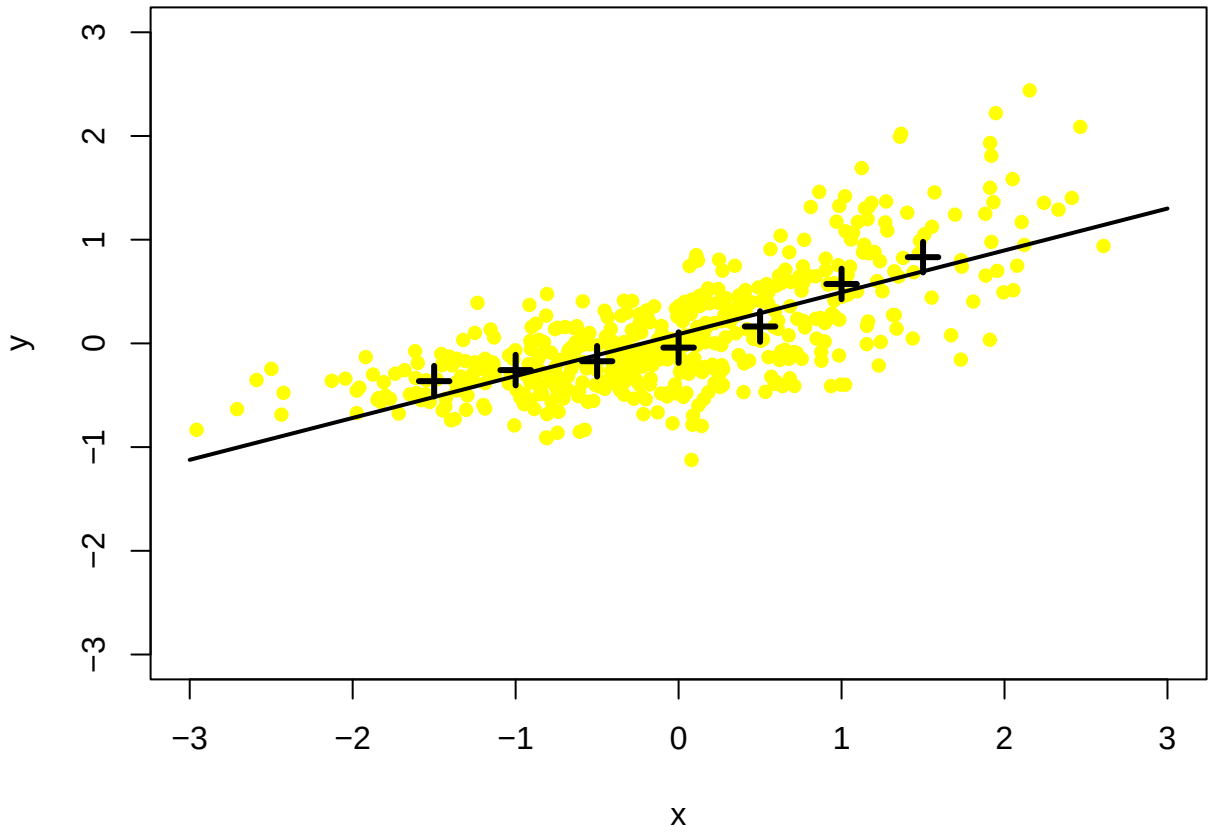

18. The graph below gives the regression line for plot C

```
graphavgregy(rtail[,1],rtail[,2])
```



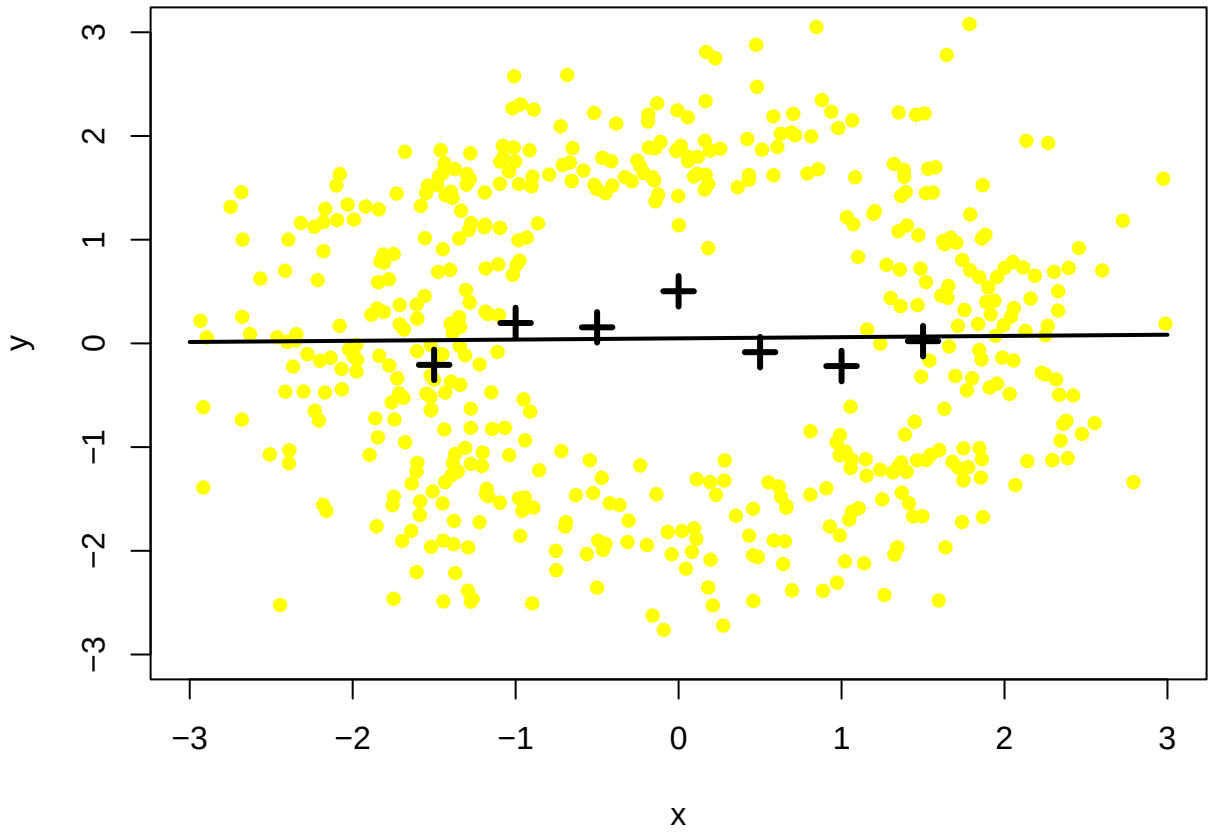

19. The graph below gives the regression line for plot D

```
graphavgregy(hetero[,1],hetero[,2])
```




20. The graph below gives the regression line for plot E

```
graphavgregy(sinusoid[,1],sinusoid[,2])
```




21a. The slope of the regression line will be $0.4 * 30/20 = 0.6$ and the line goes through the point of averages $(140, 130)$. Hence the equation of the line will be $y - 130 = 0.6(x - 140)$, or $y = 0.6x + 46$.

21b. The root mean squared error will be $\sqrt{1 - 0.4^2} * 30 = 27.5$.

21c. The predicted value will be $0.6 * 120 + 46 = 118$.

21d. The range would be $118 \pm 2 * 27.5 = 63 - 173$.

21e. The slope will be $0.4 * 20/30 = 0.27$ and the line goes through the point of averages $(140, 130)$. Hence the equation of the line will be $x - 140 = 0.27 * (y - 130)$, or $x = 0.27y + 104.9$.

21f. Solving for $x$ in terms of $y$ in the equation from part a. gives $x = (y - 46)/0.6 = 1.7y - 77.7$. This is not the same equation as in part e.

21g. If $y$ increases by 1 points, then the regression method says that $x$ should increase by $(1/30)*0.4*20 = 0.27$ points on average, but the equation in part f. predicts that $y$ will increase by 1.7 points. This does not make sense.

21h. You are assuming that the relationship between the midterm and final score is the same even if the midterm score is outside the range of midterm scores that you are basing the regression line on.

21i. This assumption could be reasonable, but turns out not to work very well in most contexts.

22. The equation of the regression line for predicting $y$ based on $x$ is $y - \bar{y} = \frac{rSD(y)}{SD(x)}(x - \bar{x}) - r(x - \bar{x})$, or $y = rx + (\bar{y} - r\bar{x})$. The equation of the regression line for predicting $x$ based on $y$ is $x = ry + (\bar{x} - r\bar{y})$. Solving for $y$ in terms of $x$ in this equation gives $y = (x + r\bar{y} - \bar{x})/r = \frac{1}{r}x + \bar{y} - \frac{\bar{x}}{r}$. The slopes of the two equations will be the same if only if $r = \frac{1}{r}$, or $r = \pm 1$. The $y$-intercepts will also be the same if $r = \pm 1$. Hence the condition for the equations to be the same is $r = \pm 1$.

23a.Let $m_1$ and $m_2$ be the slopes of the two regression lines. It follows that $m_1 = corr(x_1, y_1) * SD(y_1)/SD(x_1) = corr(x_2, y_2) * SD(y_2)/SD(x_2) = m_2$, and so the slopes of the two regression lines are the same. Similarly, the root mean squared errors are the same since $RMS_1 = \sqrt{1 - corr(x_1, y_1)^2} SD(y_1) = \sqrt{1 - corr(x_2, y_2)^2} SD(y_2) = RMS_2$.

23b. Let $m$ denote the slope of the two regression lines. The first regression line will have an equation $y - \overline{y_1} = m(x - 0)$, or $y = mx + \overline{y_1}$ and the second regression line will have an equation $y - \overline{y_2} = m(x - 10)$, or $y = mx + (\overline{y_2} - 10m)$. Hence $\overline{y_1} = \overline{y_2} - 10m$ or $\overline{y_2} = \overline{y_1} + 10m$.

23c. Let $n_1$ and $n_2$ denote the sizes of the two datasets. Suppose that the regression line for the combined dataset is different than the regression line for the individual datasets. The mean squared error on each of the two datasets will be larger with this line than with the regression line for the two datasets since the regression line minimizes the mean squared error on each of the two datasets. This means that the mean squared error on the combined dataset with this line will be larger than the mean squared error on the combined dataset with the regression line, since the mean squared error on the combined dataset is a weighted average of the mean squared errors for the individual datasets, with the weights being $w_1 = n_1/(n_1 + n_2)$ and $w_2 = n_2/(n_1 + n_2)$. This contradicts the definition of the regression line, and so the regression line for the combined dataset must be the same as the regression line for the individual datasets.

23d. We will need to know the average of the squares and the square of the average from the combined dataset in order to calculate the standard deviation. The averages of the squares from the two datasets will be $SD^2(y_1) + \overline{y_1}^2$ and $SD^2(y_2) + \overline{y_2}^2 = SD^2(y_1) + (\overline{y_1} + 10m)^2$ from parts a. and b. The combined average of squares will be a weighted average $w_1(SD^2(y_1) + \overline{y_1}^2) + w_2(SD^2(y_1) + (\overline{y_1} + 10m)^2) = SD^2(y_1) + \overline{y_1}^2 + w_2(20m\overline{y_1} + 100m^2)$ since $w_1 + w_2 = 1$. The average of the combined dataset will be $w_1\overline{y_1} + w_2\overline{y_2} = w_1\overline{y_1} + w_2(\overline{y_1} + 10m) = \overline{y_1} + 10w_2m$. Taking the difference between the average of the squares and the square of the average gives $SD^2(y_1) + \overline{y_1}^2 + w_2(20m\overline{y_1} + 100m^2) - (\overline{y_1} + 10w_2m)^2 = SD^2(y_1) + 100m^2(w_2 - w_2^2) > SD^2(y_1)$ since $w_2 < 1$. Hence the standard deviation in the combined dataset will be larger than the standard deviation in the individual datasets.

23e. The RMS error for the combined data set is the same as for the individual datasets and the standard deviation of $y$ for the combined dataset will be larger than the standard deviation of the individual datasets by part d. Hence the correlation between $x$ and $y$ in the combined dataset will be larger than the correlation in the individual datasets since $RMS_{combined} = \sqrt{1 - corr^2_{combined}} SD(y_{combined}) = RMS_1 = \sqrt{1 - corr^2(x_1, y_1)} SD(y_1)$.